FRC 2026 REBUILT — Top-30 Robot Design Deep Dive

A study reference for effective design & implementation, drawn from Chief Delphi, David's Design Server (frcdesign.org), team CAD/code releases, Behind the Bumpers, and Statbotics EPA.

The 2026 design landscape

The game

REBUILT is a fuel (6" foam ball) shooter: collect fuel and score it into your hub when active. Robots may hold unlimited fuel. The end-game tower climb existed but was LOW VALUE — most top teams dropped or never built a climber, so 2026 design is about the fuel cycle: intake → storage/indexing → launch.

The defining meta

The season split into two offensive families: (1) high-volume 'DUMPERS' — huge hoppers that drive to the hub and empty fuel in bulk during scoring windows (1678 'Limestone', 254, 7769); and (2) 'SHOOTERS' — flywheel or drum launchers, frequently TURRETED with vision auto-aim for shoot-on-the-move (4414, 6329, 1987, 2767, 5000, 4481). Several strong teams REBUILT mid-season from a passer/multi-wide shooter to a dumper (2910 Blitz→ReBlitz; 9470 triple→drum; 5687 'Big Dumpy').

Indexing was where teams innovated

Moving fuel by geometry beat squeezing it. Spindexers ('car wash' dual-stack on 1796, zero-compression drum on 6329, cone-in-bowl on 9483), drum/revolver magazines, and roller-floor serializers (1678, 1768) dominated. Large flow-through hoppers (50–80 fuel) were common.

Drivetrain & vision

Swerve is universal at this tier. Vision was a real differentiator but was rarely published: confirmed stacks include 9470 (2× OV2311 global-shutter + Orange Pi 5 Pro, shoot-on-the-move derived from 6328), 1678 (Limelight 4 + CANrange), 3467 (custom Jetson + 4 cameras), 5000 (PhotonVision), 157 (PhotonVision on Orange Pi 5). Most teams left vision hardware unstated — marked 'unconfirmed' here, meaning not published, not absent.

How to use this document

Profiles are ordered by Statbotics EPA (season-cumulative scoring contribution). For each team we capture archetype, the four subsystems, vision, public CAD/design availability, and 'lessons for students'. The teams with the richest reproducible documentation — best for a hands-on study — are 1678, 4414, 1690, 7769, 6329, 6800, 3467, 9470, 5000, and 4481. ACCURACY NOTE: subsystem specifics (motors, gear ratios, materials) are compiled from each team's public CAD, tech binder, reveal, and code, and may contain transcription or interpretation errors — always confirm a detail against the team's own linked source before relying on it. Corrections from teams are applied verbatim (e.g., Team 5000's entry was corrected on 2026-07-06 at the team's request).

Top-30 at a glance (by Statbotics EPA)

#	Team	Robot	EPA	Rec	Archetype	Public CAD/Design
1	*** 4414 HighTide**	RIPCURRENT	**357**	70-2	Single-turret flywheel shooter + 'dye-rotor' centering; high-B	Tech binder
2	**254 The Cheesy Poofs**	Overload	**328**	65-6	'Overload': 4-wide DUMPER with an articulating hood. Chose dum	Team site
3	*** 7769 The CREW**	CHUNK	**312**	70-5	Wide single-stream 'dumper' (chose dumper over turret) with a	CD thread
4	**1323 MadTown Robotics**	(undisclosed)	**310**	69-4	Unconfirmed — strong results, minimal public documentation	Stats/TBA
5	**2056 OP Robotics**	(2026 binder pending)	**302**	63-10	Single-ball turret shooter fed by a high-throughput conveyor +	Video
6	*** 1690 Orbit**	KEPLER	**295**	(district)	'Kepler' trench-bot built on 4 pillars: shoot-on-the-move, lar	CAD release

#	Team	Robot	EPA	Rec	Archetype	Public CAD/Design
7	**27 Team RUSH**	(undisclosed)	**295**	83-6	Wide 3-fuel STEEL-DRUM shooter fed by a roller floor + passive	Stats/TBA
8	*** 1114 Simbotics**	Simbot Tim	**288**	62-7	Wide DRUM dumper (deliberately NO turret); huge fuel capacity.	Team site
9	**2481 Roboteers**	(undisclosed)	**285**	44-8	'Big dumper': fixed 4-wide shooter + 5-ball-wide touch-it-own-	Stats/TBA
10	**5687 The Outliers**	Glenn / 'Big Dumpy'	**283**	62-12	Fast drum-shooter that iterated into a dumper ('Big Dumpy')	Code (GitHub)
11	*** 2910 Jack in the Bot**	Blitz → ReBlitz	**282**	53-23	Built a passer/3-wide shooter ('Blitz'), then REBUILT mid-seas	CAD release
12	**125 NUTRONs**	FLUX	**282**	68-17	'Dye rotor' ('dish') shooter - NUTRONs originated the dye-roto	Team site
13	*** 9470 Ctrl-Alt-Defeat**	(triple → drum)	**279**	60-13	Triple shooter rebuilt mid-season into a drum shooter; 'CTRL+A	Tech binder
14	**7558 ALT-F4**	(undisclosed)	**279**	51-17	Drum shooter with an advanced vision/simulation software progr	Team site
15	*** 1678 Citrus Circuits**	Limestone	**278**	61-9	World-record DUMPER / hooded shooter; led the 'Certified Dumpe	CAD release
16	**6369 Mercenary Robotics**	(full-ramp intake bot)	**274**	77-9	Rebuild from a TURRET to a wide aluminum DRUM shooter; slapdow	Code (GitHub)
17	*** 6800 Valor**	Downpour	**268**	62-10	Elevator-launched fuel scorer with assisted-aim and bidirectio	Onshape CAD
18	*** 3467 Windham Windup**	(Open Alliance)	**266**	75-10	High-throughput dumper with auto-aim shooter backup; software/	Onshape CAD
19	**9128 ITKAN Robotics**	Triple Threat	**265**	60-7	Trench robot: cycloidal slapdown intake -> duct-tape conveyor	Video
20	**1796 RoboTigers**	Sprout	**263**	48-7	'Car Wash' spindexer fuel scorer	CD thread
21	**9483 Overcharge**	Enigma	**261**	38-6	Minimalist, low-motor-count spindexer scorer	Video
22	*** 6329 The Bucks' Wrath**	Roman I / Roman II	**260**	61-15	REBUILD story: 'Roman I' (turret + low-compression spindexer,	Onshape CAD
23	**2046 Bear Metal**	Bearricade MkII	**260**	78-16	'Big dumper' with a hopper-floor + drum shooter; simplicity-ma	GrabCAD
24	**1987 Broncobots**	Cyclone	**259**	54-6	'Cyclone': turret-mounted DYE-ROTOR shooter (up to 720deg trac	Video
25	**1768 Nashoba Robotics**	Nightshift	**259**	70-17	Fixed full-width high-throughput floor-loader; volume shooter,	CD thread
26	**8044 Denham Venom**	Cat-5	**257**	50-9	Ran two robots: v1 (low 'bump'/under-trench) and v2 (a HighTid	DDS thread
27	*** 1706 Ratchet Rockers**	Mirage	**256**	45-2	Dual-turret shooter ('Mirage' - 'shooting so fast you think yo	CD thread
28	**9496 LYNK**	Matterhorn	**254**	70-13	2-wide drum 'over-bumper reverse' shooter (modeled on 254's 20	CD thread
29	**5460 Strike Zone**	Helios	**252**	57-11	Drum shooter mounted on a turret (a 'turret dumper'), ~30deg t	CD thread
30	*** 581 Blazing Bulldogs**	Rubble	**251**	47-19	'Rubble': dye-rotor turret shooter w/ 35-deg adjustable hood,	Onshape CAD

* = recommended study pick (high performance + a usable public CAD or design release).

Team-by-team profiles

#1 (study pick)	**4414 HighTide** RIPCURRENT San Diego, CA	**EPA 357** 70-2

Single-turret flywheel shooter + 'dye-rotor' centering; high-BPS single stream

SHOOTER 4× Kraken X44 on a 3" flywheel; turret geared 55.7:1 with a flex-wheel cable-chain tensioner; low gearing keeps launch current low	**INTAKE** Over-bumper/ground intake, 2× Kraken X60 + 1× X44 bottom bar; left/right sides decoupled for durability
INDEXER / HOPPER 'Dye rotor' optimized for high-BPS single-stream feeding; structural bumpers double as hopper walls AND the shooter mount	**DRIVETRAIN** 25×32" swerve, 2 custom modules from WCP Swerve X2 parts, 7.67:1; rear corners chamfered to grow hopper volume
ENDGAME Climber not detailed in the binder.	**AUTONOMOUS** High-BPS single-stream turret with vision auto-aim / shoot-on-the-move (controls-led; precomputed trajectory maps).
VISION Limelight always pointed at the hub (reads the hub's 4 AprilTags), accounting for ~95% of on-robot localization [BtB transcript]	**PUBLIC CAD / DESIGN** Interactive tech binder (no downloadable CAD)

LESSONS FOR STUDENTS

- Make structure multi-functional: bumpers serve as hopper walls and shooter mount, saving weight and volume.
- Low flywheel gearing minimizes current spikes at launch — protects the whole electrical budget.
- Decoupling the two intake sides improves survivability against contact/defense.
- A polished, AI-assisted state-machine codebase + interactive binder is a model of documentation.

Tech binder: https://2026.team4414.com/ · Tech binder · Reveal · Statbotics · Behind the Bumpers (full walkthrough)

#2	254 The Cheesy Poofs Overload San Jose, CA	EPA 328 65-6

'Overload': 4-wide DUMPER with an articulating hood. Chose dumper over turret at their 12-hr kickoff build (turret ceiling ~18 BPS vs ~30-35 for a 4-wide multi-stream); robustness-first. ~25-28 BPS on fresh balls. Curie Division winner [BtB transcript]

SHOOTER 4-wide dumper shooter with an articulating hood and a 'triangle of compression' - each ball contacts the main drum, the bottom feeder roller, and the top hood roller at once -> zero bobble, full spin-up, consistent shots. Only ~1.25in clearance per side, so a unique DOUBLE-SIDED belt drives the main drum (outer) and the back hood rollers (inner) for the reversing effect; backing feeder swapped poly -> CatTongue (poly slipped); shooter motors nested behind [BtB transcript]	**INTAKE** Most-iterated subsystem. Rack-and-pinion deploy: racks went from 3/8in polycarb w/ titanium teeth (kept breaking) to TWO layers of 1/4in SRP + one 1/8in titanium (unbreakable). A bottom 3-roller rips balls up TITANIUM kicker plates/bars (aluminum bent) tuned for low compression without dead spots; a 2nd roller keeps stored hopper balls from blocking incoming ones; dual Kraken X60. (Used GoPros inside to find jams; SRPP chosen to spread load around bolt holes, ~30% lighter, impact-resistant) [BtB transcript]
INDEXER / HOPPER 'Bumper hopper' = a full 8in polycarb wall extending up from the bumpers (more width, fewer parts) + a carbon-fiber hoop/corners for rigidity; a NET raises capacity from ~50 to ~75-80 balls. Hopper floor = WCP rubber-back timing belts (smoother than rollers). The floor's pivot is LINKED to the intake deploy ('trash compacting'): pulling the intake in while shooting lifts the floor so gravity empties the last balls; the intake fully stows under the floor [BtB transcript]	**DRIVETRAIN** Swerve (module not detailed); deliberately does NOT shoot on the move (current limits) - instead optimizes for very fast alignment + a smart pivot point [BtB transcript]
ENDGAME Climber not detailed (dumper archetype).	**AUTONOMOUS** Dumper: drive to the hub and empty fuel in bulk during scoring windows; auto routines not detailed.
VISION FOUR cameras (their most ever) on ONE Jetson Orin Nano with GPU-accelerated AprilTag detection: two grayscale 120fps on the shooter side (heavy lifting) + two 60fps side cameras; MULTI-TAG only (rejects single-tag), ~100+ fps. The Jetson solves and sends poses over UDP to the RIO, which filters by reprojection error + ambiguity. Heading control uses a predicted look-ahead pose + feed-forwards + an intelligent center-of-rotation (pivots about a corner for fastest alignment); Choreo 3-stage autos with a recovery path [BtB transcript]	**PUBLIC CAD / DESIGN** No public CAD (reveal video + build blog only)

LESSONS FOR STUDENTS

- 254's build blog shows disciplined prototyping (backspin single-wheel vs dual flywheel) before committing.
- Even the #2 team in the world withheld CAD — when a team publishes nothing, match video is your scouting source.
- Their strategic game-analysis method is as worth studying as the hardware.

Team site: https://www.team254.com/first/2026/ · 254 2026 · Reveal · DDS robot-pics-2026 · Behind the Bumpers (full transcript)

#3 (study pick)	7769 The CREW CHUNK Michigan	EPA 312 70-5

Wide single-stream 'dumper' (chose dumper over turret) with a very high-capacity hopper

SHOOTER Evolved from a two-chamber shooter to a 2.75"-wide shooter; praised for distance + accuracy	**INTAKE** Linear slapdown intake; deploy arms cut from polycarbonate rack + 98A TPU printed chain tensioner; dynamic current limit (45A deploy → 5A hold) for compliance
INDEXER / HOPPER Very high capacity — ~70 fuel max, 50–60 in play; in/out throughput was their key edge	**DRIVETRAIN** Swerve (module unconfirmed)
ENDGAME Climber not detailed (dumper).	**AUTONOMOUS** Drive-and-empty dumper cycle; auto routines not detailed.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** Public CAD + tech slides (Onshape)

LESSONS FOR STUDENTS

- Massive hopper capacity + high throughput beat turret complexity for them — a clear dumper case study.
- Candid failure analysis: polycarb intake teeth sheared every 2–3 matches at champs; they ran ~16 spares. Pick intake materials for fatigue.
- Software current-limit tuning can buy you mechanical compliance without extra hardware.

CD thread: https://www.chiefdelphi.com/t/frc-7769-cad-tech-slides-chunk/521008 · CAD + slides · Behind the Bumpers

#4	1323 MadTown Robotics (undisclosed) Madera, CA	EPA 310 69-4

Unconfirmed — strong results, minimal public documentation

SHOOTER Unconfirmed	**INTAKE** Unconfirmed
INDEXER / HOPPER Unconfirmed	**DRIVETRAIN** Swerve (assumed)
ENDGAME Unconfirmed (minimal public documentation).	**AUTONOMOUS** Unconfirmed (minimal public documentation).
VISION Unconfirmed	**PUBLIC CAD / DESIGN** No public CAD/design release

LESSONS FOR STUDENTS

- Strong results (#4, 69-4) don't require public documentation — design knowledge isn't always shareable.
- When a team publishes nothing, use TBA match video as the primary scouting source.
- A candidate to contact directly rather than study from public CAD.

Stats/TBA: https://www.thebluealliance.com/team/1323/2026 · Team site · TBA 2026

#5	2056 OP Robotics (2026 binder pending) Stoney Creek, ON	EPA 302 63-10

Single-ball turret shooter fed by a high-throughput conveyor + ~62-ball net hopper; world-record scorer

SHOOTER Turret-mounted single-ball shooter: 2x Kraken X60 flywheel run at ONE fixed RPM (never varied up/down), hood on one Kraken X44; 370deg rotation via a sprung, non-bidirectional cable chain [BtB transcript]	**INTAKE** Linear EXTENDING intake with Lexan flex side-plates (survives collisions, enters opponents' zone); front rollers on a Kraken X60 with plain-rubber coating; surgical-tubing kicker bar added mid-season to stop balls sticking against the bumper; 3D-printed wire-management chain [BtB transcript]
INDEXER / HOPPER Neoprene-coated 'hot dog' rollers + a live-bottom funnel; a bare-Lexan 'slip' roller before the conveyor as an anti-jam; conveyor = top+bottom timing belts (thicker WCP bottom belt) compressing balls up into the turret; a reverse-spinning anti-jam roller; net hopper (~62 fuel) repurposed from prior years [BtB transcript]	**DRIVETRAIN** Trapezoidal swerve base (per Behind the Bumpers)
ENDGAME Climber not detailed in the public material.	**AUTONOMOUS** Single-ball turret + high-throughput conveyor; auto/vision detail in their Behind the Bumpers video.
VISION Not named in the Behind the Bumpers walkthrough (throughput is conveyor-driven with a single-speed flywheel); camera/coprocessor unconfirmed	**PUBLIC CAD / DESIGN** 2025 binder public; 2026 not yet released

LESSONS FOR STUDENTS

- 2056's 'Behind the Bumpers' series is a model for student-facing design explanation — watch the 2026 entry.
- Their prior technical binders (e.g. 2025) are an excellent template for a student design study.
- Research trap to avoid: 'LIGHTNING' is their 2025 robot, not 2026 — verify season before citing.

Video: https://www.youtube.com/@2056public · Team site · YouTube · TBA 2026 · Behind the Bumpers (read full transcript)

#6 (study pick)	**1690 Orbit** KEPLER Israel	**EPA 295** (district)

'Kepler' trench-bot built on 4 pillars: shoot-on-the-move, large (~60-ball) hopper, extreme durability, low profile; roller-floor indexer into a 640deg turret. Galileo Division winner [BtB transcript]

SHOOTER Compact 640deg turret (360+ per side): X60 drive through a low gearbox (low CG), self-made gears on an 8in bearing, ball path offset toward the back to tuck into the corner. Shooter tuned for slight/no spin (most consistent stream, lowest energy loss): back rollers belted to a custom main shooter wheel; motors on an SLS winch pulled tight to the wheel + hood motors on the plate side via bevel gears to MINIMIZE diameter; custom 3D-printed energy chain + 14ga wires + a raised 'neck' so the chain wraps tight enough for 640deg [BtB transcript]	**INTAKE** Durable/collapsible/swappable wide intake: two silicone-coated rollers on a Kraken X60 + a static bottom kicker roller; deliberately NO slapdown (poor impact collapse) - instead an in-house RACK-AND-PINION deploy (polycarb racks + aluminum/brass gears); 25x50mm front extrusion + a polycarb/aluminum side sandwich; the whole U-frame lifts off via 3 screws per side for fast swaps [BtB transcript]
INDEXER / HOPPER 3 stages: (1) roller floor of 11 carbon-fiber silicone rollers on an X60 (2:1), roller-to-roller via 3mm GT2 belts, a slight reduction to ease the 4-wide -> 3.5-wide jam point, plus two slick (un-coated) rollers to cut friction; (2) two 30mm 5mm-HTD vertical belts on two independent X44 (~8 m/s) over a 1mm PTFE sheet (~10mm compression) + a vertical kicker roller (repurposed from their 2024 Doppler shooter) on an X44 to clear jams; (3) an active 'C' shooter ramp (3 rows of 2.25in flex wheels + 2 rows of 2in stealth wheels) on an X60 -> 8 m/s with ZERO spin into the turret [BtB transcript]	**DRIVETRAIN** 29.5x24.5in (wide side = intake); bumpers attach to the hopper walls (thin polycarb + a net for ~60-ball capacity). MID-SEASON swap from a custom MK5-based module to stock SDS MK5 modules after the defense bot pushed them around - the MK5's wide spike treads dig into the carpet vs their thin nitro treads; reinforced/integrated top plates. 'Defense mode' X-locks the swerve [BtB transcript]
ENDGAME Low-profile 'trench-runner' geometry trades climb access for faster, safer collection — climbing is not a focus.	**AUTONOMOUS** Shoot-on-the-move is one of the four stated design pillars (see CAD/reveal).
VISION A camera ON THE TURRET on an Orange Pi running PhotonVision (aimed at the hub's AprilTags) + two rear cameras for blind spots when the turret cable wraps or a camera drops; tilt + bump detection lets it shoot while tilted; an off-robot shoot-on-the-move simulation (gravity, air resistance, Magnus) precomputes trajectory polynomials the robot evaluates live [BtB transcript]	**PUBLIC CAD / DESIGN** Full public CAD (Onshape + x_t)

LESSONS FOR STUDENTS

- Choosing a low-profile/trench geometry trades climb access for faster, safer ball collection.
- First full year on Onshape — a good case study in CAD-tool migration and clean release practice.
- A full-floor roller intake maximizes pickup area and tolerates sloppy approaches.

CAD release: https://www.chiefdelphi.com/t/frc-orbit-1690-2026-robot-cad-release/520673 · CAD release · Reveal 'Kepler' · Video · Behind the Bumpers (full transcript)

#7	27 Team RUSH (undisclosed) Clarkston, MI	EPA 295 83-6

Wide 3-fuel STEEL-DRUM shooter fed by a roller floor + passive hopper; high-throughput dumper. Hall of Fame; Michigan State champion; ~330 EPA, 113 lb [BtB transcript]

SHOOTER One main STEEL 3in drum run by 4x Kraken X60 + two 1in hood rollers on the same X60s (spin ~2x faster to minimize backspin); 1x X44 rack-and-pinion hood; mostly plastic SRP + carbon-fiber structure for weight. VARIABLE compression (aluminum rods ~5/8in, CF stability rods ~3/8in, final ~1/2in) to fit a retrofit shooter compactly; the feeder accelerates fuel BEFORE the shooter, so the shooter only adds the final boost [BtB transcript]	**INTAKE** Main roller on 2x Kraken X44 (2.5:1) + a free-spinning carbon-fiber top roller + 3 passive inner tubes; built for quick fixes (it's at the robot's fighting face); the HOPPER is driven PASSIVELY off the intake [BtB transcript]
INDEXER / HOPPER Roller floor + indexer on 2x Kraken X60; floor = 1in polycarb rollers (5 with CatTongue tape) funneling to the 3-wide shooter via dead-axle 3D-printed guides; geared 1:1 in front, ~17:19 at the back to speed balls up; a free-spinning roller prevents jams (replaced a belted one). Iterated from belt/'sushi'/chain to all-polycarb rollers [BtB transcript]	**DRIVETRAIN** Swerve; ALL electronics mounted on the bottom (frees space for the roller floor + shooter); the electrical cover attaches to the BUMPERS so its weight doesn't count against inspection; 3D-printed component mounts [BtB transcript]
ENDGAME Unconfirmed — reveal is announcement-only.	**AUTONOMOUS** Unconfirmed — reveal is announcement-only.
VISION Main Limelight 4 (heavily weighted, accepts tags mainly from the rear given the shooting orientation) + left & right Limelights fused as one entity into the pose estimate for odometry during feeding / when the back can't see [BtB transcript]	**PUBLIC CAD / DESIGN** No public CAD/design release

LESSONS FOR STUDENTS

- An 83-6 record (highest win count in the top-30) with no public release: execution and consistency can outweigh documentation.
- Reveal video + match footage are the only primary design sources.
- Michigan's competitive density (27, 7769, 5460...) is itself a design driver — strong regions force strong robots.

Stats/TBA: https://www.thebluealliance.com/team/27/2026 · Reveal thread · Video · TBA · Behind the Bumpers (full transcript)

#8 (study pick)	1114 Simbotics Simbot Tim St. Catharines, ON	EPA 288 62-7

Wide DRUM dumper (deliberately NO turret); huge fuel capacity. Hall of Fame; won both Ontario districts + the Ontario provincial championship [BtB transcript]

SHOOTER Big belt-driven DRUM, an X44 + 3 more motors; hood uses only ~8deg of travel. They chose a drum (more versatile) OVER a turret (didn't suit their play style) and chose not to go under the trench [BtB transcript]	**INTAKE** 2x Kraken X60; balls hit a kicker bar + main roller; an 'agitating' intake with multiple states - it can vibrate fast to push balls into the indexer harder, or run gentler [BtB transcript]
INDEXER / HOPPER Extending hopper -> a moving 'rolling floor' (two rollers) carries balls up to the drum [BtB transcript]	**DRIVETRAIN** Swerve (not detailed in the walkthrough)
ENDGAME Climber not detailed (wide drum dumper, no turret).	**AUTONOMOUS** Drive-and-empty dumper; auto routines not detailed.
VISION 3x Limelight cameras with MegaTag2 + AprilTags, fused with odometry for pose estimation [BtB transcript]	**PUBLIC CAD / DESIGN** No 2026 CAD release confirmed

LESSONS FOR STUDENTS

- Simbotics is well-known for STRATEGIC DESIGN EDUCATION — their resources/presentations are a top study reference even without 2026 CAD.
- Watch the 'Simbot Tim' Behind the Bumpers for subsystem rationale.
- Pair with 2056 (also Ontario) to compare two strong programs' 2026 approaches.

Team site: https://www.simbotics.org/resources · Showcase · Behind the Bumpers · Resources

#9	**2481 Roboteers** (undisclosed) Tremont, IL	**EPA 285** 44-8

'Big dumper': fixed 4-wide shooter + 5-ball-wide touch-it-own-it intake + 115-ball hopper (no turret); Milstein Division winners [BtB transcript]

SHOOTER 4-wide fixed shooter, ~20 BPS; 4 accelerator rollers for accuracy; 3x Kraken X44 on the shooter + 2x X44 + 1x X60 on feed rollers; grip-tape (skateboard tape) for consistent ball spin (+~40% accuracy), a late-season PVC-pipe guide (+~10%); 2-position linear-actuator hood (replaced a pneumatic piston to cut weight), flywheel RPM varied for distance [BtB transcript]	**INTAKE** 5-ball-wide 'touch-and-take' intake: 3in roller wrapped in grip tape on 2x Kraken X60, SRPP side plates with 'Flintstone' wheels; ball compressed between roller and ramp; anti-jam = a 2nd motor + a plate to stop the roller stalling [BtB transcript]
INDEXER / HOPPER Massive 115-ball, 5-wide hopper funneling to the 4-wide shooter through the accelerator rollers [BtB transcript]	**DRIVETRAIN** Custom swerve (designed in the 2024 offseason) with very large wheels and an adjustable ratio (12-32 ft/s); geared LOW this year for torque/low current under the heavy fuel load [BtB transcript]
ENDGAME Climber not detailed (fixed 4-wide dumper, no turret).	**AUTONOMOUS** Dumper; auto routines not detailed.
VISION Not mentioned in the Behind the Bumpers walkthrough; camera/coprocessor unconfirmed	**PUBLIC CAD / DESIGN** No public 2026 release (GitHub stops at 2025)

LESSONS FOR STUDENTS

- Strong record (44-8) with a low public footprint — another reminder strong results needn't be open-sourced.
- Study via match video; consider contacting the team directly.

Stats/TBA: https://www.thebluealliance.com/team/2481/2026 · Team site · TBA 2026 · Behind the Bumpers (full transcript)

#10	**5687 The Outliers** Glenn / 'Big Dumpy' Portland, ME	**EPA 283** 62-12

Fast drum-shooter that iterated into a dumper ('Big Dumpy')

SHOOTER 'Insanely quick' near-full drum shooter; high cycle rate prioritized	**INTAKE** Over-bumper roller intake — 2" upper + 1.25" lower roller with ~½" compression (minimized to avoid bogging the motor)
INDEXER / HOPPER CTRE/Kraken 'Feeder' subsystem feeding a calculated-shot ('ShotCalculator') shooter; 3-roller intake (top/bottom rollers + deployer)	**DRIVETRAIN** Swerve (team has historically experimented with differential swerve)
ENDGAME Climber not detailed (drum -> dumper).	**AUTONOMOUS** Fast drum/dumper cycle; auto routines not detailed.
VISION Hybrid Limelight + PhotonVision: 3 cameras (front + left + right), AprilTag pose (MegaTag); C++ codebase	**PUBLIC CAD / DESIGN** Code public (GitHub, C++); no CAD release

LESSONS FOR STUDENTS

- Empirical compression tuning — they minimized roller compression by experiment to reduce motor load.
- A separate 'LifeBot/Big Dumpy' repo signals a deliberate mid-season pivot to the dumper meta.
- A C++ codebase (less common than Java) is a good study for advanced programmers.

Code (GitHub): https://github.com/frc5687 · 'Glenn' reveal · GitHub · TBA · Robot code (C++)

#11 (study pick)	2910 Jack in the Bot Blitz → ReBlitz Mill Creek, WA	EPA 282 53-23

Built a passer/3-wide shooter ('Blitz'), then REBUILT mid-season into a lightning dumper ('ReBlitz')

SHOOTER Blitz: 3-fuel-wide shooter, whole robot aims like a 3-wide turret, ~30 fuel/sec; ReBlitz dumper hit 30+ balls/sec	**INTAKE** Dual intakes on Blitz
INDEXER / HOPPER Blitz used a multilayered indexer + intentionally SMALL hopper (a noted tradeoff); ReBlitz went to an open dumper path	**DRIVETRAIN** Swerve (long in-house swerve lineage)
ENDGAME Climber not detailed (mid-season rebuild to a dumper).	**AUTONOMOUS** Dumper after the Blitz -> ReBlitz rebuild; auto routines not detailed.
VISION Unconfirmed for 2026	**PUBLIC CAD / DESIGN** Onshape CAD + tech binders for BOTH robots (CD release, 2026-06-07)

LESSONS FOR STUDENTS

- THE case study in the season's central lesson: they scrapped a working passer and rebuilt as a dumper — and it won PNW DCMP + a return to Einstein.
- Now fully studyable: downloadable Onshape CAD AND a technical binder were released for BOTH Blitz and Re-Blitz, so you can diff the original vs the rebuild directly.
- Explicitly shows the hopper-capacity vs throughput tradeoff.
- Demonstrates the courage/value of abandoning a working robot mid-season when the meta shifts.

CAD release: https://www.chiefdelphi.com/t/2910-cad-and-tech-binder-release-2026/521705 · CAD + Tech Binder release · Blitz reveal · Behind the Bumpers

#12	125 NUTRONs FLUX Boston, MA	EPA 282 68-17

'Dye rotor' ('dish') shooter - NUTRONs originated the dye-rotor concept (2017) and went BIGGER this year; trapezoidal drive base [BtB transcript]

SHOOTER Enlarged dye-rotor 'dish' (sized up this year for throughput) fed by a live-shaft 3in carbon-fiber drum (CatTongue tape + silicone wrap) with a 3-standoff kick bar that maintains ball compression through the path [BtB transcript]	**INTAKE** Four-bar linkage deploy, modular (whole intake pops off via 2 shoulder bolts per side); a stiffener bar for aggressive driving; the intake is carved out so the robot can sit UNDER the trench at match start [BtB transcript]
INDEXER / HOPPER Live-shaft 3in CF drum + 3-standoff kick bar feeding the dye-rotor dish [BtB transcript]	**DRIVETRAIN** Trapezoidal base - wider at the FRONT (bigger intake) and narrower at the REAR (balls rarely reach the back); 1/4in side plates (no 2x1 main structure); 2 stock WCP Swerve X2S modules at the front + 2 custom-machined X2S at the rear, shaped to the dish radius [BtB transcript]
ENDGAME Climber not detailed (dye-rotor shooter).	**AUTONOMOUS** Dye-rotor shooter; auto routines not detailed.
VISION Not mentioned in the Behind the Bumpers walkthrough; camera/coprocessor unconfirmed	**PUBLIC CAD / DESIGN** No 2026 CAD release located

LESSONS FOR STUDENTS

- A 68-17 record from a distributed 5-school co-op is a model for sustaining strong performance across a large org.
- Mechanism detail lives in the FUN 'Behind the Bumpers' episode — watch rather than read.

Team site: https://www.nutrons.com/ · Reveal thread · Behind the Bumpers · TBA

#13 (study pick)	9470 Ctrl-Alt-Defeat (triple → drum)	EPA 279
	Half Moon Bay, CA	60-13

Triple shooter rebuilt mid-season into a drum shooter; 'CTRL+ALT+DEL' deadline-driven scope control

SHOOTER Drum shooter (mid-season rebuild from a Week-1 triple shooter): open fuel path, three feeder rollers on two Kraken X60.	INTAKE Reinforced beater-bar intake — a flippable 2x1 aluminum bar (flip it when bent), then two 0.25\" SRP leaf-spring plates per side with the intake tube as a hardstop and a detachable roller mount.
INDEXER / HOPPER Hopper with COTS aluminum hubs + compliant floor wheels carried over from the triple-shooter robot.	DRIVETRAIN Swerve; a CANivore drives 18-motor shoot-on-the-move bursts.
ENDGAME Climber not detailed — engineering hours went to scoring and intake robustness.	AUTONOMOUS Empirical shot map (measured, not modeled; first-shot +250 RPM compensation from data) with shoot-on-the-move.
VISION Two OV2311 global-shutter cameras + Orange Pi 5 Pro coprocessors; SOTM / vision / current-limiting adapted from 6328 and the NINJAS; 4.6 V brownout floor so failures degrade gracefully.	PUBLIC CAD / DESIGN Excellent public tech binder + GitHub

LESSONS FOR STUDENTS

- Their 'CTRL+ALT+DEL' rule ships the robot 1.5 weeks before every event — the deadline decides what to cut, copy, or skip (applied 3x: build, rebuild, reinforce).
- Reinforce the proven failure point (the intake) instead of chasing optimization — robustness over speed for playoffs.
- Agentic coding tools for boilerplate let juniors plan instead of writing Java from scratch.

Tech binder: https://9470techbinder.vercel.app/ · Tech binder · Engineering process · Behind the Bumpers

#14	7558 ALT-F4 (undisclosed)	EPA 279
	North York, ON	51-17

Drum shooter with an advanced vision/simulation software program; rebuilt after their 2nd event; Ontario Technology Division champions [BtB transcript]

SHOOTER 3in aluminum DRUM with silicone tubing + two 1.5in silicone hood rollers; drum powered by 4x Kraken X60 geared 1:1; plus a custom laser-cut 1/2in-thick x 4in-wide STEEL flywheel [BtB transcript]	INTAKE Linear intake running on HTD parts with 10DP gear-tooth racks meshing 10DP gears [BtB transcript]
INDEXER / HOPPER Unconfirmed	DRIVETRAIN Swerve (not detailed in the walkthrough)
ENDGAME Climber not detailed.	AUTONOMOUS Advanced vision/simulation software program drives auto-aim; routines in code.
VISION Multiple Limelights with a sophisticated localization pipeline: per-camera outlier rejection -> 'smart' MegaTag1/MegaTag2 selection via hysteresis -> staged standard-deviation calc; logs everything to simulate the robot without owning a field [BtB transcript]	PUBLIC CAD / DESIGN No public 2026 release located

LESSONS FOR STUDENTS

- Their 2024 turret-on-swerve + efficient diverter suggests strong mechanism-aiming experience.
- 2026 design is video-only — treat as a watch-the-reveal study.

Team site: https://www.team7558.com/ · Team site · YouTube · TBA · Behind the Bumpers (full transcript)

#15 (study pick)	**1678 Citrus Circuits** Limestone Davis, CA	**EPA 278** 61-9

World-record DUMPER / hooded shooter; led the 'Certified Dumper Group' with 254

SHOOTER Hooded flywheel fed from a full-width 3.5" aluminum drum (grip-taped); hood on an X44 jackshaft for a wide shot-angle range	**INTAKE** Full-width slapdown ground intake — a ~6 lb aggressive over-bumper roller (Kraken X60 pivot, silicone roller + kicker bar)
INDEXER / HOPPER Roller floor (dead-axle + flex-wheel rollers) → ball tunnel/serializer; expanding hopper; pragmatic 'strategically placed wood' guides	**DRIVETRAIN** Swerve, SDS MK5n, 27×27"
ENDGAME Climber not detailed (world-record dumper / hooded shooter).	**AUTONOMOUS** Drive-and-empty dumper; auto routines not detailed.
VISION Limelight 4 (single rear unit) + 2× CANrange sensors; custom localization + a driver dashboard with a 'shift' system (CONFIRMED)	**PUBLIC CAD / DESIGN** Full CAD + code (Onshape, GitHub)

LESSONS FOR STUDENTS

- The reveal that 'confirmed the dumper meta' — a community thread analyzed what Limestone implied about REBUILT strategy.
- Localization + a custom driver dashboard show software/driver UX as a force multiplier.
- Pragmatic 'wood in the ball path' — strong teams still use simple, cheap fixes.
- Fully open CAD + code makes this the single best reproducible study target of the season.

CAD release: https://www.chiefdelphi.com/t/1678-citrus-circuits-2026-cad-and-robot-code-release/521535 · Reveal: Limestone · Onshape CAD · Behind the Bumpers

#16	**6369 Mercenary Robotics** (full-ramp intake bot) McKinney, TX	**EPA 274** 77-9

Rebuild from a TURRET to a wide aluminum DRUM shooter; slapdown kicker-ramp intake -> rolling floor -> drum; net hopper (~78 fuel). Won Amarillo/Waco/San Antonio + Impact + EI [BtB transcript]

SHOOTER 3in aluminum DRUM (only ~1/6in wide -> too little momentum, so they added a 1.5 lb WCP STAINLESS flywheel at 1:3 - stainless beat brass, which wore down); drum on 4x Kraken X60 at 1:1; shooter geared 60:1 on a single X44 (down from 400:1); transfer = two 2in + one 1.25in rollers tilted back to passively funnel; back rollers two 1.25in silicone; ~20 ft/s for 60-70 balls; all motors packaged under the drum. Chose drum over turret for higher throughput + simpler/more-reliable software [BtB transcript]	**INTAKE** Slapdown intake (kept from robot 1) with a full KICKER RAMP (vs a kicker bar) for max fuel contact time + sustained compression (Citrus-style push into the indexer); 3 bars lift fuel high so the rolling floor can empty the hopper without much agitation [BtB transcript]
INDEXER / HOPPER Rolling floor of custom 3D-printed stub rollers linked by belts (the intake jackshaft is hidden inside a roller to use the full width; one roller pivots with the intake); hopper extends exactly 12in + a net -> ~50-55 fuel un-expanded, up to ~78 at 30in with the net up (for bump-overfill cycles) [BtB transcript]	**DRIVETRAIN** Swerve - swapped WCP X2 -> SDS MK5N (easy to replace), geared for speed + torque [BtB transcript]
ENDGAME Climber not detailed (turret -> drum rebuild).	**AUTONOMOUS** Auto routines not detailed; chose drum for simpler, more-reliable software.
VISION Three Limelight 4s (two front, one back) giving full-field localization in teleop + auto; one-button point-and-drive aiming with X-lock against defense; auto-aim auto-adjusts during autos [BtB transcript]	**PUBLIC CAD / DESIGN** Code on GitHub; no confirmed 2026 CAD

LESSONS FOR STUDENTS

- Best win-rate in the cohort (77-9) despite a lower EPA ceiling — a case study in consistency and alliance leadership over raw scoring.
- 'Full ramp intake' points to a wide, forgiving ground pickup optimized for reliability.

Code (GitHub): https://github.com/mercs6369 · Behind the Bumpers · GitHub · DDS robot-pics-2026

#17 (study pick)	**6800 Valor** Downpour Austin, TX	**EPA 268** 62-10

Elevator-launched fuel scorer with assisted-aim and bidirectional shooting

SHOOTER Dual carbon-fiber arm launcher fed up a ~47" elevator (~0.3 s travel); can shoot in two directions	**INTAKE** Rotating intake synchronized with the arm for hand-off
INDEXER / HOPPER Internal feeder routing fuel intake→launcher; designed for fast maintenance access	**DRIVETRAIN** Swerve
ENDGAME Climber not detailed.	**AUTONOMOUS** Assisted-aim, bidirectional elevator-launched scorer; auto routines not detailed.
VISION Top-mounted camera for assisted scoring/alignment with a reverse-kinematic software model (published in their tech packet)	**PUBLIC CAD / DESIGN** Public CAD (Onshape)

LESSONS FOR STUDENTS

- One living CAD assembly iterated all season (branches per version) keeps design history in one place.
- Maintenance-driven layout (top electronics, bottom battery plate, ziptie battery-lead hooks) — design for the pit, not just the field.
- An elevator-launch architecture is a creative alternative to flywheel/turret shooters.
- Published a full Tech + Engineering-Ops binder — strong documentation discipline.

Onshape CAD: https://cad.onshape.com/documents/15cd41462dc24ff986a4e549 · Tech binder (PDF) · CAD release thread · Reveal · Behind the Bumpers

#18 (study pick)	**3467 Windham Windup** (Open Alliance) Windham, NH	**EPA 266** 75-10

High-throughput dumper with auto-aim shooter backup; software/autonomous-led

SHOOTER Primarily a bulk dumper (fill hopper, empty during on-shifts) with an auto-aim shot vs tower/trench	**INTAKE** Linear-slide retracting intake with polycarbonate hopper walls; tested 1- vs 2-motor intake
INDEXER / HOPPER Large hopper + indexer/tower rollers; tuned indexer floor angle and speed for throughput	**DRIVETRAIN** Swerve on Thrifty Narrow modules (switched from MK4i), CANivore/CANcoder, Choreo paths
ENDGAME Climber not detailed.	**AUTONOMOUS** Software/autonomous-led: a bulk dumper with an auto-aim shooter backup.
VISION Custom NVIDIA Jetson + 4-camera setup (CONFIRMED); AprilTag localization	**PUBLIC CAD / DESIGN** Full public CAD (Onshape) + Open Alliance blog

LESSONS FOR STUDENTS

- Autonomous CONSISTENCY beats peak score — they were passed at alliance selection for inconsistent autos, then fixed odometry (lowered Choreo friction coeff, restored drivetrain current limits).
- New-carpet and CAN-terminator gremlins caused real losses — document environmental variables.
- Their Open Alliance blog is a model of transparent post-event self-critique.
- Deliberate 'simple week-1 bot, planned mid-season upgrade' scheduling.

Onshape CAD: https://cad.onshape.com/documents/7ef2c49c5bb48212a6b180ee · Open Alliance blog · CAD

#19	9128 ITKAN Robotics Triple Threat Plano, TX	EPA 265 60-7

Trench robot: cycloidal slapdown intake -> duct-tape conveyor -> transfer -> drum shooter, with a passive 'hopper flopper'. Won Space City 1 + Farmersville + an Impact award [BtB transcript]

SHOOTER Drum shooter: 2x Kraken X60 on a ~13 lb flywheel mass (steel flywheels + wheels); BANG-BANG control (from their FTC shooter experience) to ramp fast, an operator pre-ramp button while crossing the bump, and a VARIABLE transfer speed (slows/speeds vs the flywheel velocity-vs-setpoint so each ball takes max energy); ~25-30 BPS [BtB transcript]	**INTAKE** Flip-down SLAPDOWN intake on a CYCLOIDAL pivot drive (Kraken X60 -> cycloidals; chosen for ease of use + better ball compression than a linear intake); two rollers (3in hard wheels + a roller bar) on 2x Kraken X44; a pin-and-slot links the intake raise to pull the hopper back in (agitation) [BtB transcript]
INDEXER / HOPPER 7-roller conveyor with DUCT-TAPE flaps to fling balls into the transfer (the tape jumped BPS from ~18-20 to ~25-30); transfer = three 1in rollers + bottom stationary tubes for compression, rollers on slots for tuning. Passive 'hopper flopper' - a flexible hopper that collapses to go under the trench and re-expands to store ~80 fuel [BtB transcript]	**DRIVETRAIN** Swerve (not detailed); built as a trench robot [BtB transcript]
ENDGAME Climber not detailed.	**AUTONOMOUS** Drum shooter; auto routines not detailed (Impact award).
VISION A single Limelight facing the shooting direction; planned a 2nd but used software instead - an auto 'point-towards' the opposing hub to update odometry before crossing the bump (so it doesn't hit the trench) [BtB transcript]	**PUBLIC CAD / DESIGN** No public CAD (reveal + Behind the Bumpers)

LESSONS FOR STUDENTS

- A drum (revolver-style) shooter trades flywheel spin-up for very high BURST rate — strong for empty-the-hopper volleys.
- Distinguish BURST throughput (20–25 BPS) from SUSTAINED (15–16 BPS) when specifying a shooter; they are different targets.
- Running two near-identical robots (9128/10340) multiplies build & test learning.

Video: https://www.youtube.com/watch?v=5HtLpv4AtlE · Reveal 'Triple Threat' · Behind the Bumpers

#20	1796 RoboTigers Sprout Long Island City, NY	EPA 263 48-7

'Car Wash' spindexer fuel scorer

SHOOTER Launcher fed from a spindexer (launch type unconfirmed)	**INTAKE** Intake pre-fills/'primes' the feeder while the outer spindexer stack turns slowly
INDEXER / HOPPER 'Car Wash' spindexer — two independently driven stacks, each on its own Kraken; outer stack spins slow during intake to evenly pack the hopper	**DRIVETRAIN** Swerve (specifics unconfirmed)
ENDGAME Climber not detailed.	**AUTONOMOUS** Auto routines not detailed ('Car Wash' spindexer).
VISION Unconfirmed	**PUBLIC CAD / DESIGN** No public CAD (photos on request)

LESSONS FOR STUDENTS

- Independently driven indexer stages (separate motors) let you intake and prime simultaneously without jamming.
- Slow-spin-while-intaking evenly distributes fuel — a simple control trick for hopper packing.
- Cited by peers as 'one of the best robots in the world' despite no public CAD.

CD thread: https://www.chiefdelphi.com/t/robotigers-team-1796-presents-sprout-2026/516769 · Reveal 'Sprout'

#21	**9483 Overcharge** Enigma Istanbul, Türkiye	**EPA 261** 38-6

Minimalist, low-motor-count spindexer scorer

SHOOTER Launcher fed from a spindexer (type unconfirmed)	INTAKE Two top rollers + a bumper-level mechanism (deliberately simple, no obvious kickers/ramps)
INDEXER / HOPPER Spindexer 'bowl' with a central cone wrapped in cat-tongue grip tape to fling fuel outward to the feeder	DRIVETRAIN Swerve (specifics unconfirmed)
ENDGAME Climber not detailed.	AUTONOMOUS Minimalist, low-motor spindexer; auto routines not detailed.
VISION Unconfirmed	PUBLIC CAD / DESIGN No public CAD (reveal video)

LESSONS FOR STUDENTS

- Reputation for 'doing things with fewer motors than anyone thinks possible' — fewer actuators means fewer failure modes, lighter/cheaper robots.
- A grip-taped cone in a rotating bowl can index fuel with almost no dedicated hardware — elegant minimalism.

Video: https://youtu.be/tOYFbRIJ8CA · Reveal 'Enigma'

#22 (study pick)	**6329 The Bucks' Wrath** Roman I / Roman II Bucksport, ME	**EPA 260** 61-15

REBUILD story: 'Roman I' (turret + low-compression spindexer, ~8 BPS) -> 'Roman II', a WIDE aluminum DRUM shooter at ~23 BPS with a net hopper (~60-70 fuel). World-high single-match score [BtB transcript]

SHOOTER Roman II: wide aluminum DRUM (custom 3D-printed segments over an ordered tube, CatTongue-taped) + an adjustable hood; flywheel speed and hood angle scale with distance (further = faster + hood up); ~23 BPS (vs ~8 on the Roman I turret). A safety routine auto-lowers the hood entering the trench, with a bigger safety zone the faster it crosses [BtB transcript]	INTAKE Custom silicone wheels (numbers printed) + an agitator linking all the rollers, CatTongue-taped for grip; a compact redesign over a wider v1 (a long bar that kept bending on hits was removed; flexible polycarb instead of Lexan to absorb impact); custom 3/8in-hex stub rollers instead of full-length shafts to save weight [BtB transcript]
INDEXER / HOPPER A net hopper raised capacity from ~35 to ~60-70 fuel; rollers + agitator feed CatTongue-taped rollers up into the drum (the whole top net/cover flips up for easy access to the battery, PDH, and motors) [BtB transcript]	DRIVETRAIN Swerve; built for serviceability (the top flips up to reach the battery / PDH / intake & swerve motors); prototyped on a wooden chassis to place electronics before final build [BtB transcript]
ENDGAME Climber not detailed.	AUTONOMOUS Vision-aimed turret; a safety routine auto-lowers the hood entering the trench (bigger zone the faster it crosses).
VISION THREE Limelights tracking AprilTags (added a 3rd on the shooter over the previous two side cameras for more accurate pose); pose-based controls - one button locks onto the target / passing mode, and an operator button auto-aligns even if pose drops [BtB transcript]	PUBLIC CAD / DESIGN Public CAD — both versions (Onshape)

LESSONS FOR STUDENTS

- Zero-compression indexing (move fuel by geometry/rotation, don't squeeze it) reduces jams and wear.
- A simple 'upkicker' (static bar + side wheels) is a low-part-count hand-off from spindexer to turret.
- Released TWO full CAD versions (Roman I & II) — a great study in in-season iteration.

Onshape CAD: https://cad.onshape.com/documents/014716f840fdbd882c1d691f · Reveal 'Roman' · Behind the Bumpers · Roman II CAD

#23 2046 Bear Metal Bearricade MkII

Maple Valley, WA

EPA 260

78-16

'Big dumper' with a hopper-floor + drum shooter; simplicity-maximizing design. PNW District Champion, World Finalist [BtB transcript]

SHOOTER 4 lb, 4in-wide aluminum DRUM coated in CatTongue tape, ~3000 rpm; 24-25 BPS average (peaks 29-30), 4x Kraken X60; X44 hood with NO rollers (rollers hurt accuracy); hood -10deg to 60deg (60deg = bump-to-bump pass; -10deg shoots BACKWARD against the hub under defense while locking the wheels) [BtB transcript]	**INTAKE** Collector on 2x Kraken X60; two sets of CatTongue rollers + a 3rd 'pop-up' roller; stub axles for weight/durability; NO front beater bar - 1/2in polycarb + aluminum plates at contact points for resilience without the weight; an upper TPU-mounted protector roller keeps balls in the hopper [BtB transcript]
INDEXER / HOPPER Hopper extends 11.5in (kept under 12in so flex doesn't breach frame perimeter), double-fold polycarb top for rigidity + a net for capacity; ~70-80 fuel full, 50-60 while still fitting under the trench. Hopper bed = 11 rollers at a 21:20 ratio (back roller faster for grip), belts ~0.03in loose to cut current; floor pivots up for battery access. S-shaped (vs J-shaped) superstructure with a single 2-Kraken feed roller for throughput + less jamming + more hopper volume [BtB transcript]	**DRIVETRAIN** Swerve (AdvantageKit-style logging of swerve states + odometry) [BtB transcript]
ENDGAME Climber not detailed (big dumper, simplicity-max).	**AUTONOMOUS** Drive-and-empty dumper; auto routines not detailed.
VISION Vision pose updates fused with swerve odometry (logged); specific camera/coprocessor not named in the walkthrough (2025 code used PhotonVision + OV9782 AprilTag cameras) [BtB transcript]	**PUBLIC CAD / DESIGN** No confirmed 2026 release (GrabCAD history)

LESSONS FOR STUDENTS

- 'MkII' naming reflects a deliberate v1→v2 redesign — iterate on a known-good base rather than restarting.
- Known for fabrication quality and wire management — clean execution scales to deep playoff runs.
- Pre-champs EPA peaked ~301, one of the highest all season — a peak-performance benchmark.

GrabCAD: https://grabcad.com/frc.2046.bear.metal-1 · TBA 2026 · Behind the Bumpers

#24 1987 Broncobots Cyclone

Lee's Summit, MO

EPA 259

54-6

'Cyclone': turret-mounted DYE-ROTOR shooter (up to 720deg tracked) fed by a four-bar full-length intake; shoot-on-the-move, ~18 fuel/sec. Heartland Regional [BtB transcript]

SHOOTER Low-profile (4.25in tall) to clear the trench. 'Floating hood' on a cycloidal 80.5:1 + a 10 lb constant-force spring for rigidity; main flywheel = 4in WCP urethane 60A on a DEAD axle, 2x Kraken X60 (24:30 ratio), deliberately NO added flywheel mass so RPM changes instantly for shoot-on-move; separate 1in 45A backspin wheels on their own powertrain (1x X60); hood on a Kraken X44, up to 90deg; partial-depth pocketing for weight [BtB transcript]	**INTAKE** Four-bar FULL-LENGTH intake (holds up to 5 balls per intake); 5-roller system - the 5th roller adds velocity to push balls deep and clear dead zones; aluminum caps keep the polycarb rollers on their hubs through hits; a free-spinning 'Zamboni wheel' lets it run along the wall without being ripped off; motors hidden down low for packaging [BtB transcript]
INDEXER / HOPPER DYE ROTOR (centering): a big outer diameter + a thin inner column; driven COAXIALLY (rotor speed independent of the wheel, tuned in code) via #25 chain (one motor wheels, one rotor); the rotor's metal rods carry much of the turret weight on two X-contact bearings; ~2.5 rot/sec. The 720deg turret runs #25 chain + a MAXPlanetary 25:1 with a 2:1-geared absolute encoder [BtB transcript]	**DRIVETRAIN** Swerve (its module velocities feed the shoot-on-the-move solver) [BtB transcript]
ENDGAME Climber not detailed.	**AUTONOMOUS** Turret dye-rotor (tracked up to 720deg) with shoot-on-the-move.
VISION THREE Limelights for real-time pose. Shoot-on-the-move predicts a FUTURE position (swerve velocity x pose vector, scaled by the ~1s ball flight time) and aims there; the inertia-free dead-axle flywheel lets RPM jump (e.g. 50->20 RPS) instantly. Intake power auto-scales by drive direction to save current [BtB transcript]	**PUBLIC CAD / DESIGN** No public CAD (Behind the Bumpers)

LESSONS FOR STUDENTS

- Turret wiring is hard: v1 over-built the umbilical; v2 ('C.U.R.S.E.D.' concealed umbilical retraction) routed wires down one protected conduit — 'less is more'.
- Minimal superstructure (one tube) keeps the fuel path to the rotor clear — let the game-piece flow path drive the frame.

Video: https://www.youtube.com/watch?v=FbkE2kAgKQE · Reveal 'Cyclone' · Behind the Bumpers

#25 1768 Nashoba Robotics Nightshift

Bolton, MA

EPA 259

70-17

Fixed full-width high-throughput floor-loader; volume shooter, NO turret

SHOOTER Full-width hooded drum shooter, 4× Kraken X60; 1/16" aluminum drum surfaced with CatTongue tape; 3D-printed hood segments between 1/16" plates; separate Kraken X60 entry roller	**INTAKE** 4-bar intake (roller plates, side hopper plates, upper links in SRPP); static kicker bars + 2× Kraken X60 rollers; deploy off a MaxSpline shaft under the floor
INDEXER / HOPPER Roller floor on WCP dead-axle stubs; belt shortened 0.030" to cut current; moving hopper sits under trench height; stretchy net gives ~65 fuel	**DRIVETRAIN** Swerve (modules unconfirmed)
ENDGAME Climber not detailed (fixed full-width, no turret).	**AUTONOMOUS** Volume shooter; auto routines not detailed.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** No public CAD (detailed reveal writeup); GitHub

LESSONS FOR STUDENTS

- You don't need a turret to be competitive — a full-width fixed shooter + huge hopper wins on throughput.
- SRPP and dead-axle construction keep a wide superstructure light and stiff.
- Tiny tuning (0.030" belt shortening) measurably reduced current draw — budget current deliberately.
- A compliant net hopper trades rigidity for capacity (~65 fuel).

CD thread: https://www.chiefdelphi.com/t/team-1768-nashoba-robotics-2026-robot-reveal-nightshift/519386 · Reveal 'Nightshift' · Behind the Bumpers

#26 8044 Denham Venom Cat-5

Denham Springs, LA

EPA 257

50-9

Ran two robots: v1 (low 'bump'/under-trench) and v2 (a HighTide/Denham 'tide clone'); dye-rotor shooter [per 8044 team members on DDS]

SHOOTER Dye-rotor shooter, belt-driven (HighTide-style); v1 had belt skip only when jammed, v2 fixed it by tensioning the belt rather than adding 10DP gear stages [per 8044 on DDS - medium confidence]	**INTAKE** Unconfirmed (2025 robot noted for a clean intake-to-handoff path — likely carryover)
INDEXER / HOPPER Dye-rotor fed (rotor flings fuel to the launcher) [per DDS photos]	**DRIVETRAIN** Swerve (specifics unconfirmed)
ENDGAME Climber not detailed (ran v1 under-trench + v2).	**AUTONOMOUS** Auto routines not detailed.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** No public CAD (DDS robot-pics photos)

LESSONS FOR STUDENTS

- A 50-9 record with little public documentation: execution + driver practice can outrank publicity.
- Their history of slick mechanism handoffs is worth studying for reliable ball transfer.

DDS thread: https://discord.com/channels/1120162219502608426/1479491912305217637 · DDS robot-pics-2026 · TBA · DDS robot-pics-2026 thread (8044)

#27 (study pick) 1706 Ratchet Rockers Mirage

Wentzville, MO

EPA 256

45-2

Dual-turret shooter ('Mirage' - 'shooting so fast you think you're seeing double') fed by a single spindexer; Oklahoma Regional winner [BtB transcript]

SHOOTER TWO turrets, each with a front kicker + a back kicker; 6in aluminum flywheels (made by sponsor Custom Machine Works - tried steel, aluminum won); hoods rotate fully [BtB transcript]	**INTAKE** Full-width four-bar intake, 2x Kraken X60 (one for pitch, one for the rollers); 4 silicone rollers (3 carbon-fiber + 1 polycarbonate, all wrapped in white silicone) for grip; the intake doubles as a hopper extension to minimize moving parts [BtB transcript]
INDEXER / HOPPER A spindexer (re-iterated several times over the season) ramps fuel up into each shooter; kicker wheels fire the front + back kicker on each turret [BtB transcript]	**DRIVETRAIN** Swerve (odometry fused with vision in the pose estimator) [BtB transcript]
ENDGAME Telescoping-tube climber (Thrifty).	**AUTONOMOUS** Dual-turret very high fire-rate ('Mirage'); auto routines not detailed.
VISION 3x Limelight 4 + swerve odometry in the pose estimator; the Limelight hardware-manager Kalman filter rejects extraneous poses and only trusts MegaTag2 on good solutions; Limelights sit under each shooter + on the climber [BtB transcript]	**PUBLIC CAD / DESIGN** Onshape CAD (champs config 'Mirage') + code on GitHub

LESSONS FOR STUDENTS

- A 45-2 record is a case study in consistency over flash — minimize failure modes, maximize cycle reliability.
- Watch their Behind the Bumpers for how a veteran team packages a clean, serviceable robot.
- Champs-config CAD is now public on Onshape — dig into the dual-turret flywheel shooter and single-spindexer feed.

CD thread: https://www.chiefdelphi.com/t/522276 · CAD release (Mirage, champs) · Behind the Bumpers · Code (GitHub) · TBA 2026

#28	**9496 LYNK** Matterhorn Spindale, NC	**EPA 254** 70-13

2-wide drum 'over-bumper reverse' shooter (modeled on 254's 2017 'Misfire')

SHOOTER 2-wide drum shooter, fixed 34deg hood (polycarb + foam, ~0.675in compression); 4 motors on one shaft (SDS brass + TTB-0106 flywheels, ~5.2 lb); flywheel RPM scaled by distance	**INTAKE** Roller floor with 'pulsing' (reverse-intake while shooting to empty the hopper)
INDEXER / HOPPER Roller floor: indexer rollers 0.375in, top floor roller 0.5in; indexer/floor/shooter run 1:1 at 12V	**DRIVETRAIN** MK5N-style swerve, ~26x26in, Spike grip wheels
ENDGAME Climber not detailed.	**AUTONOMOUS** 2-wide drum 'over-bumper reverse' shooter; auto routines not detailed.
VISION Single AprilTag camera for pose (primarily off the hub tags); a 2nd camera under consideration	**PUBLIC CAD / DESIGN** Open Alliance build log + GitHub

LESSONS FOR STUDENTS

- A young team (3rd year) reaching #28 EPA shows the value of an Open Alliance build log — public iteration accelerates learning.
- Out-of-state district scheduling (VA→GA→SC) is a program-building lesson, not just a robot one.

CD thread: https://www.chiefdelphi.com/t/lynk-9496-build-log-2026/511797 · Open Alliance build log · Reveal 'Matterhorn'

#29	**5460 Strike Zone** Helios Lapeer, MI	**EPA 252** 57-11

Drum shooter mounted on a turret (a 'turret dumper'), ~30deg total turret travel

SHOOTER Drum shooter on a ~30deg-travel turret	**INTAKE** Unconfirmed
INDEXER / HOPPER Unconfirmed	**DRIVETRAIN** Swerve (historically SDS MK4i)
ENDGAME Climber not detailed (drum-on-turret 'turret dumper').	**AUTONOMOUS** Auto routines not detailed.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** 2026 release pending (strong tech-binder history)

LESSONS FOR STUDENTS

- Strike Zone habitually publishes a full tech binder (2023, 2025) — review those as a documentation template.
- A team known for 'accurate' shooters underscores investing in shot consistency over raw range.

CD thread: https://www.chiefdelphi.com/t/5460-strike-zone-2026-reveal/515689 · 2026 'Helios' reveal · 2025 CAD + binder

#30 (study pick)	581 Blazing Bulldogs Rubble San Jose, CA	EPA 251 47-19

'Rubble': dye-rotor turret shooter w/ 35-deg adjustable hood, pivoting floor intake, WCP X2i/X2c swerve

SHOOTER Dye-rotor turret ('floating turret' de-conflicts rotor table vs turret). Drum: 4 in OD x 1/16 in-wall aluminum wrapped in Cat Tongue grip tape, powered by four Kraken X60s. Adjustable hood with 35 deg of rotation on a Kraken X44 at 1:120 through a zombie roller; two 2 in Cat Tongue hood rollers geared up 1.64:1 plus three 1.5 in silicone-sleeved rollers on two X60s (1:1.5). Chose the short dye-rotor turret over a large dumper after week-1 Alpha Bot testing hit ~15 BPS but lacked consistency.	**INTAKE** Pivoting floor intake: pivots up/down for bellypan access and ball compaction; elastic X-pattern springing keeps constant tension so balls don't jam under the floor, and the front roller is lowered so balls can't cam under during the compaction sequence. Custom Onyx-printed pulleys with 1/16 in steel dowels to prevent layer shearing; installs onto the shooter tower with two bearing plates.
INDEXER / HOPPER Floor rollers feed the shooter tower. A CANrange measures hopper fill and auto-runs the rollers until a retroreflective sensor in the tower trips - buffering ~8 extra fuel in the tower without ejecting prematurely.	**DRIVETRAIN** 28.5 x 27 in frame (1/8 in-wall tube, 5/8 in corner chamfers to stay under the 110 in perimeter). Two WCP X2i + two X2c swerve modules, X1 gearing with 10T pinions, Kraken X60 drive / X44 steer; custom 1/16 in bottom plates to drive over the Depot, intake-side module top plates cut down for a lower intake.
ENDGAME Climber not detailed.	**AUTONOMOUS** Dye-rotor turret with adjustable hood (chosen over a dumper after Alpha-Bot testing); auto routines not detailed.
VISION 3x Limelight 4 (rear 'SHOOTER' + left + right), AprilTag MegaTag2; Limelight-only pipeline (no PhotonVision)	**PUBLIC CAD / DESIGN** Full robot CAD (Onshape) + code (GitHub, team581/frc-2026)

LESSONS FOR STUDENTS

- Week-1 'Alpha Bot' archetype test: they built a quick dumper, measured ~15 BPS but poor consistency, and let DATA kill the archetype before committing - cheap insurance against a season-long mistake.
- Software power manager: a separate thread re-allocates current limits across subsystems in real time (e.g. prioritize the shooter while scoring) to prevent brownouts over a full match.
- Defensive autonomy: if all cameras drop, the robot tells the drivers and falls back to pre-set shots; dashboard toggles disable complex features mid-match.
- Turret wire management without e-chain: cable sleeving + surgical tubing for tension - low-cost and light.

Onshape CAD: https://cad.onshape.com/documents/766d37b113335d7a60cc2375/w/8b35fc383279c1954cec5b72/e/bd3188f47c23702bef9f1fc8 · CAD + Code release (CD) · Reveal (full Q&A) · Code (GitHub)

Additional well-documented robots for study (beyond the EPA top-30)

These rank below the EPA top-30 but each has a strong public release (CAD, code, binder, or detailed reveal) and a clear design lesson. Ordered by EPA. ★ marks the recommended study picks.

#8 (study pick)	**1114 Simbotics** Simbot Tim Ontario, Canada	**EPA 288** 62-7

Wide DRUM dumper (deliberately NO turret); huge fuel capacity. Hall of Fame; won both Ontario districts + the Ontario provincial championship [BtB transcript]

SHOOTER Big belt-driven DRUM, an X44 + 3 more motors; hood uses only ~8deg of travel. They chose a drum (more versatile) OVER a turret (didn't suit their play style) and chose not to go under the trench [BtB transcript]	**INTAKE** 2x Kraken X60; balls hit a kicker bar + main roller; an 'agitating' intake with multiple states - it can vibrate fast to push balls into the indexer harder, or run gentler [BtB transcript]
INDEXER / HOPPER Extending hopper -> a moving 'rolling floor' (two rollers) carries balls up to the drum [BtB transcript]	**DRIVETRAIN** Swerve (not detailed in the walkthrough)
ENDGAME A climber was designed and appears in the CAD, but per the team it was added on March 1 and removed by March 19 for weight, and stayed off the rest of the season as the game developed.	**AUTONOMOUS** Not described in the release thread - see the CAD/code if released.
VISION 3x Limelight cameras with MegaTag2 + AprilTags, fused with odometry for pose estimation [BtB transcript]	**PUBLIC CAD / DESIGN** 2026 CAD Release (Onshape)

LESSONS FOR STUDENTS

- The active-vs-passive roller split in the hopper is a concrete fix for game-piece bridging ('crystallizing'): one driven floor roller (Kraken X44 via a cross-robot belt) agitates the stack while the top rollers stay passive - a low-cost way to keep fire rate consistent.
- Intake was refined over many iterations in collaboration with 1678 - a good example of cross-team prototyping focused on a single subsystem.
- A top-8, 62-7 Hall-of-Fame robot with public CAD is a strong benchmark for overall packaging and mechanism integration.
- Endgame pragmatism: the climber was cut mid-season on a weight budget and never re-added - a reminder that a designed mechanism has to earn its weight.

Onshape CAD: https://cad.onshape.com/documents/7959d159451177786d52dd9a/w/535bc31b66b0cc442967f1d0/e/a263c1a625ed8cf2484ae82f · CAD Release (CD) · CAD (Onshape) · Statbotics 1114 (2026) · The Blue Alliance 1114 (2026)

#34 (study pick)	**5000 Hammerheads** Icarus Marblehead, MA	**EPA 236** 41-23

Turreted vision-tracking fuel scorer; full Open Alliance build + open-source code

SHOOTER Turret-mounted flywheel shooter with adjustable hood; camera-tuned auto-aim. Shot parameters (hood angle + wheel speed) come from InterpolatingTreeMap lookup tables with empirically-measured time-of-flight. Turret weight/reduction detail unconfirmed.	**INTAKE** Rack-and-pinion deploy driven by Kraken X44 motors (on TalonFX controllers), ~48:17 & 50:17 belt reductions; spin rollers
INDEXER / HOPPER Dye rotor plus a spin motor and dual feed motors (feed + follower); unjam logic with hook/feed stall detection	**DRIVETRAIN** Swerve (CTRE, Kraken motors on TalonFX controllers)
ENDGAME Climber not detailed.	**AUTONOMOUS** Shoot-on-the-move: 5 iterations of time-of-flight recursion with velocity- and latency-compensated turret tracking; robot auto-starts scoring ~2s before an active shift and stops ~1s after. Full autonomous routines are in the code / tech binder.
VISION 6x Arducam OV9281 on PhotonVision (front L/R/center + back L/R/center), MegaTag; welded-field AprilTag layout used at Worlds	**PUBLIC CAD / DESIGN** Championship CAD (Fusion 360) + open GitHub code

LESSONS FOR STUDENTS

- Well-documented shoot-on-the-move controls: InterpolatingTreeMap shot lookup tables, empirical time-of-flight, and velocity/latency-compensated turret tracking — a concrete controls case study (see the Icarus 2.0 tech binder).
- Full Open Alliance build thread + public Fusion 360 CAD + open GitHub code (incl. a TurretIOTalonFX subsystem) document a turret robot end-to-end — an approachable scope for smaller teams.

Team site: https://a360.co/4wuUiK0 · Championship CAD (Fusion 360) · Tech binder (Icarus 2.0) · Open Alliance build thread · GitHub (2026Rebuilt)

#36 (study pick)	**359 Hawaiian Kids** Poi Pounder XXVII (V1 & V2) Hawaii, USA	**EPA 233** 69-11

Turret + adjustable-hood flywheel shooter, slapdown intake, two-stage tunnel+accelerator indexer, sliding hopper, swerve, climber; V1 & V2 CAD + 28-page binder

SHOOTER Adjustable-hood flywheel on a turret. Flywheel: 2x Kraken X60 (1:1) spinning three 4" 60A Stealth wheels + two trimmed SDS 4" brass flywheels (added moment of inertia), with five 1" Sushi accelerator rollers and a 3D-printed back ramp to compress the FUEL. Hood: Kraken X44 (167.5:1), 3D-printed sector-and-pinion, 28 deg range. Turret: 7.5" ID X-Contact bearing, custom 8ST/10DP gear, +/-200 deg (400 deg total) on a Kraken X44 (50:1), biased to the back-left corner for shoot-on-the-move.	**INTAKE** Motor-powered slapdown, three full-width horizontal rollers, touch-it-own-it. Deploy: Kraken X44 (50:1) via #25 chain on a coaxial shaft (also drives the floor belts) plus a spring-loaded kicker bar; the deployment motion also actuates the prismatic sliding hopper. Rollers: Kraken X60 (1.60:1), 1.5" polycarb tubes in silicone; 1/4" pocketed 7075 side plates, SRPP front guard. (V2 rebuild swapped this slapdown for a linear intake whose plates directly carry the extending hopper.)
INDEXER / HOPPER Two stages. Tunnel (Kraken X60, 1.88:1): top + bottom conveyors, two 15 mm belts each, align the FUEL into one controlled stream. Accelerator (Kraken X44, 1.46:1): three 2" Stealth + custom-lathed 2.5" Colson wheels fling the FUEL up into the turret. Fed by a sliding/extending hopper and a floor conveyor (back kicker roller -> Kraken X60, 6.30:1).	**DRIVETRAIN** SDS Mk5n R3 swerve (4 modules), Kraken X44 drive + steer, 26.09:1 steering, 4" molded spike-grip wheels (2.25" wide), 19.2 ft/s. 28x26" pocketed aluminum frame; Pigeon 2.0 IMU + CTRE CANcoders on all four modules.
ENDGAME Has a climber (per the 28-page binder); part of the V1->V2 design.	**AUTONOMOUS** Single-driver, two-button automation: auto-aim + auto spin-up by field position, hood auto-lowers under the trench; PathPlanner autos; 5-camera AprilTag localization; shoot-on-the-move.
VISION Five cameras feeding full-field AprilTag localization (robot pose, vision targets, and trajectories all modeled in physics simulation), logged via AdvantageKit / AdvantageScope 3D replay; specific camera brand not named in the binder. Drives single-driver, two-button stateless automation: auto spin-up + auto-aim to the hub or an alliance-zone corner by field position, with the hood auto-lowering under the trench; PathPlanner autos with a profiled point-to-point PID controller for precise climbs.	**PUBLIC CAD / DESIGN** CAD for both robots (Onshape) + 28-page technical binder

LESSONS FOR STUDENTS

- A two-stage indexer — a 'tunnel' that single-streams the FUEL, then an 'accelerator' that adds exit velocity — cleanly decouples organizing balls from launching them.
- Stateless, two-button automation (one intake, one shoot) let them run a SINGLE driver: the robot auto-aims and auto-spins by field position and auto-lowers the hood under the trench.
- A mid-season rebuild (V1 slapdown + rail-carriage hopper -> V2 linear intake + directly-coupled hopper) is packaging-driven iteration; both versions' CAD are public for an A/B study.
- Validating every subsystem in physics-based simulation (FUEL, vision targets, and trajectories modeled) sped iteration and de-risked hardware before the robot existed.

CAD release: https://www.chiefdelphi.com/t/team-359-hawaiian-kids-2026-cad-release-q-a/521575 · CD CAD release + Q&A · Statbotics · The Blue Alliance

#32 (study pick)	694 StuyPulse HAL New York, USA	EPA 229 47-12

Swerve (WCP x2i) turret-shooter: hooded single-flywheel, ground intake, vertically-expanding hopper, L1 side-hang climber.

SHOOTER Hooded single-flywheel shooter mounted on a turret. Pre-season prototyping compared double-flywheel and single-flywheel overhand/underhand shooters; they built a hooded single-flywheel plate set with a polycarb sheet curved around the plates and a slot to adjust top-roller compression. The released Onshape model ('HAL 2026') has a 'Shooter + Turret' subassembly (33 features); wheel type, motors, and ratios are in the CAD, not stated verbatim in the thread.	**INTAKE** Ground intake; the team reused their 2025 coral ground intake for early testing and planned a wider intake to enlarge the hopper footprint. CAD has 'Intake' and 'Conveyor' subassemblies. Roller geometry/motors: see the CAD.
INDEXER / HOPPER Conveyor feeds fuel from the intake to the turret; the CAD 'HAL 2026' has 'Indexer' (8) and 'Conveyor' (4) features. The team's design goal was to intake and shoot simultaneously via the turret. Detailed geometry lives in the CAD.	**DRIVETRAIN** Swerve on WCP x2i modules, raised using the module's top-mount style (tubing sandwiched above the main plate with a WCP top plate) to gain ~1in clearance over the field bump; Kraken drive motors. The alpha/Kitbot used older MK4i modules from past robots.
ENDGAME L1 side-hang climber (CAD 'Climber', 8 features). The team deprioritized climbing versus hopper capacity but kept an L1 climb to preserve the endgame ranking point; an L3 climb was judged worth only a fraction of a cycle in this game.	**AUTONOMOUS** Not detailed in the thread; the turret was chosen partly for more time-efficient autos (shoot while intaking regardless of heading). See the code repo.
VISION Limelight (the CAD includes 'Limelight Views'; the team's plan stated adding a Limelight). Pipeline details unconfirmed; see the CAD/config.	**PUBLIC CAD / DESIGN** 2026 CAD (Onshape) - HAL

LESSONS FOR STUDENTS

- Turret trade study: 694 documented the pros/cons of a turret (shoot and ferry independent of drivetrain heading, keep the hopper full while intaking) against the added complexity and wire management, and chose to build one - a clear worked example of the decision.
- Vertically self-expanding hopper: the lid stays below trench height (about 22in) when empty and passively rises toward the 30in height limit as fuel fills the robot, adding a layer of storage without blocking trench access - a specific packaging idea worth studying.
- WCP x2i top-mount raise: mounting the tube above the module's main plate with a WCP top plate raised the whole chassis for about 1in of bump clearance without lifting the frame off the floor.
- Climb cost-benefit: they treated the L1 climb as a ranking-point insurance policy rather than a scoring priority after estimating the climb was worth only a fraction of a cycle.

Onshape CAD: https://cad.onshape.com/documents/2b44ae58b051bc8b70f03c29 · Open Alliance build thread (CD) · 2026 CAD - HAL (Onshape) · Code (GitHub, StuyPlus-2026) · Statbotics 694 (2026) · The Blue Alliance 694 (2026)

#36	1833 Team BEAN Clover Georgia, USA (Peachtree)	EPA 225 60-8

Laned-shooter robot with a kicker-plate feed and a floor intake; full Onshape CAD + public code released. Drivetrain type not stated in the release thread.

SHOOTER Laned shooter (the team ran laned shooters rather than a drum). Before Worlds they weighed rebuilding from laned shooters to a drum and decided to keep the lanes, citing time, budget, and a judgment that other subsystems and driver practice would yield more. Wheel type, hood geometry, motors and gear ratios are not stated in the thread; read them from the Onshape CAD / code.	**INTAKE** Floor intake, reworked before Worlds. The team states that prior to Champs they were unable to intake from the depot, so they invested in the intake ahead of Newton. Roller count/geometry and motors are not described in the thread; inspect the Onshape CAD.
INDEXER / HOPPER Kicker-plate feed moves fuel to the shooter. The team notes they had issues with the strength of the kicker plates before Worlds. Detailed geometry is not in the thread; see the CAD.	**DRIVETRAIN** Not stated in the release thread. Drivetrain unconfirmed here (do not assume); see the Onshape CAD.
ENDGAME Not described in the release thread; see the CAD/code.	**AUTONOMOUS** Won the Autonomous Award (Google) at both Gainesville and the Peachtree District Championship. Specific routines are not detailed in the thread; see the code repo.
VISION Not mentioned in the release thread; unconfirmed. Check the CAD/code.	**PUBLIC CAD / DESIGN** CAD (Onshape)

LESSONS FOR STUDENTS

- A documented case of NOT rebuilding: 1833 weighed switching from laned shooters to a drum before Worlds and instead spent the time on intake and driver practice. A useful example of scoping late-season changes to expected payoff.
- Intake was the limiting subsystem going into Champs (they could not intake from the depot) and was the focus of their pre-Worlds work.
- A top-40 (local rank 36), 60-8 district powerhouse and Newton Division (Worlds) Finalist with downloadable CAD + code is a strong overall packaging reference.
- Two-time Autonomous Award winner: the public code repo is worth reading for autonomous structure.

Onshape CAD: https://cad.onshape.com/documents/11b1fc83786708b81b063e5c/w/a7854643bff51f67974cba0b/e/82ecb4d9374be4387f3eecba · CAD + code release (CD) · CAD (Onshape) · Code (GitHub) · Statbotics 1833 (2026) · The Blue Alliance 1833 (2026)

#55	8214 Cyber Unicorn China	EPA 217 16-8

Ultra-clean sheet-metal turret shooter; #1 EPA in China, championship build quality

SHOOTER Custom CNC-machined flywheel (bore geometry designed to be cut with a 2.5mm end mill); turret-mounted shooter	**INTAKE** Roller intake (motors unconfirmed)
INDEXER / HOPPER Indexer/hopper present (capacity not stated in thread)	**DRIVETRAIN** Swerve (module unconfirmed)
ENDGAME Climber not detailed in the thread.	**AUTONOMOUS** Auto routines not detailed.
VISION Unconfirmed (not detailed in CAD thread)	**PUBLIC CAD / DESIGN** Full CAD release (Onshape)

LESSONS FOR STUDENTS

- Design parts AROUND your tools (DFM): they shaped the flywheel bore so a 2.5mm end mill could cut it.
- Outsource sheet metal to a sponsor, machine the rest in-house - that division of labor builds a clean, precise robot fast.
- Tidy wire management and packaging correlate with reliability and top EPA - visual order is an engineering signal.

CAD release: https://www.chiefdelphi.com/t/frc-cyber-unicorn-8214-2026-champ-robot-cad-release/521310 · CD champ-robot CAD release · Statbotics

#56 (study pick)	**4481 Team Rembrandts** (Open Alliance) Eindhoven, NL	**EPA 214** 37-15

Precision turreted hooded-flywheel scorer; strong controls

SHOOTER Turret hooded flywheel - their 4TH shooter design: a static back wall (they tried twin rollers but found them unneeded for high BPS), Teflon tape to cut friction/backspin, a custom flywheel to hold RPM, and a switch to METAL gears (the 3D-printed ones damaged easily) for champs. (Underlying: dual NEO Vortex single axle, ~370deg travel, herringbone rack-pinion hood 15-45deg) [BtB transcript]	**INTAKE** PASSIVE LINEAR intake (slides in/out, doesn't fold): NO deploy motors - constant-force springs + gravity drop it at match start; can't retract mid-match (deemed unnecessary); very impact-resistant since it can fold in on a front hit without breaking [BtB transcript]
INDEXER / HOPPER 'Washing machine' spindexer: spins very fast; iterated to a LARGER outer chamber + inner cone (helps balls settle -> higher BPS) with the gearbox + two motors packaged INSIDE the cone; lowered the washing-machine walls to admit balls easier. Turret cable chain kept taut by an elastic band (shortest at the anchor, stretches as it spins - never snags) [BtB transcript]	**DRIVETRAIN** Swerve (not detailed in the walkthrough)
ENDGAME Climber not detailed.	**AUTONOMOUS** Precision turreted hooded-flywheel with vision auto-aim; strong controls.
VISION Strong vision system: FOUR cameras at different angles to approximate robot pose; heavy focus on power management (per-subsystem current limits + supply voltage) [BtB transcript]	**PUBLIC CAD / DESIGN** Public CAD (GrabCAD) + Open Alliance thread

LESSONS FOR STUDENTS

- Excellent gear-ratio documentation (45.7:1 flywheel, 58.5:1 hood) — a model for sizing reductions to control-vs-speed needs.
- A through-bore encoder on the hood enables true closed-loop angle control, not open-loop presets.
- Their turret cable-sleeve fix (lengthening it to go from ~270° to ~370°) — design the wire path for full range from the start.
- Authored the widely-used '3DM' game-analysis document.

GrabCAD: https://grabcad.com/team.rembrandts.4481-1 · Open Alliance thread · GrabCAD · Behind the Bumpers (full transcript)

#62	**3476 Code Orange** Sandspit Irvine, CA	**EPA 212** 54-9

'Sandspit': four-bar intake -> pleated-belt roller floor -> full-width DRUM shooter (NO turret). Flow-obsessed; ~15 BPS. Orange County District [BtB transcript]

SHOOTER Full-width DRUM shooter (~15 BPS) with two massive ~5in in-house-machined flywheels (steel compound, mass concentrated at the outer radius for max inertia-per-weight, cut on a Haas mini-mill over a multi-day, multi-tool process); active hood on an in-house 10DP gear, symmetrical with a shaft + a 1:1 through-bore encoder for shoot-and-move; 3D-printed hood backings [BtB transcript]	**INTAKE** Four-bar intake with active ball control (its linear push turns the intake into a ramp funneling fuel toward the shooter); a 'scotch yoke' makes the hopper box follow the intake up/down; custom-machined 6061 sprockets (stock shortage); pivot driven through a 1in round tube to both ends. Tried coaxial roller power (balls caught) -> switched to Citrus's double-motor intake (much better) [BtB transcript]
INDEXER / HOPPER A 2:1 gearbox drives the roller floor + the first feeder roller; 'pleated belts' (TPU-printed cleats) on 3D-printed CF crown rollers agitate + push balls linearly down the ramp (replacing two top rollers that let balls fall back); crown belts drive the whole floor without belts between every roller (less friction). An 'epoxy coupler' roller mount (epoxy + embedded 4-40 nuts) cuts ~50% of stub-roller weight + allows quick roller swaps [BtB transcript]	**DRIVETRAIN** Swerve (not detailed publicly)
ENDGAME Climber not detailed (focus is on ball flow).	**AUTONOMOUS** Auto routines not detailed.
VISION Switched from Limelight to PhotonVision: SIX cameras - four AprilTag (two forward, two back) + two rear OBJECT-DETECTION cameras that find the densest fuel; 1600x800 (sharper than Limelight) with a custom ML algorithm that crops each frame to the AprilTags so it computes only on them (keeps ~50 fps + a very stable pose); two Rubik Pi 3 co-processors [BtB transcript]	**PUBLIC CAD / DESIGN** Reveal video only (no downloadable CAD)

LESSONS FOR STUDENTS

- Treat ball flow as THE design objective - anything a ball can sit on, wedge above, or spin on is the enemy.
- Dividers/lanes that look organized often trap balls; smooth continuous paths beat segmented ones.
- Make the feeder transparent and film with high-speed cameras to diagnose flow problems you can't see at speed.
- Use the intake/box as a compactor to push fuel into the feeder while shooting.

CD thread: https://www.chiefdelphi.com/t/team-3476-code-orange-2026-sandspit-robot-reveal/516875 · CD reveal + flow philosophy · Reveal video · Behind the Bumpers

#71 (study pick)	**195 The CyberKnights** Big Daddy V2 'Biggest Daddy' Southington, CT	**EPA 206** 45-21-1

Over-bumper slapdown intake -> moving floor -> uptake -> DRUM shooter (swapped from a 4-lane shooter mid-season); custom ROS2 software. New England DCMP [BtB transcript]

SHOOTER 4in TITANIUM drum + a 1:1 BRASS flywheel + two carbon-fiber back rollers at 2:1; ~25 BPS. Started as a 4-lane shooter (~15 BPS) but balls jammed on the lanes -> removed the lanes and went to the drum [BtB transcript]	**INTAKE** Over-bumper SLAPDOWN deployed off an X60 (planetary 9:1 + two 3:1 for a very fast deploy, esp. in auto); 3 polycarb rollers - a bottom kicker (no silicone; it caught on the bump) + two silicone front rollers; one Kraken X60 ~3:1. Swapped from an earlier rack-and-pinion + 4-roller intake that stuck/broke on hits (new one lighter + more robust) [BtB transcript]
INDEXER / HOPPER Moving floor = WCP stub-roller system (carbon-fiber rollers with silicone) on one Kraken X60 (~1:1, pushes balls hard into the uptake); uptake = six 1in carbon-fiber silicone tubes on 2x Kraken X60 [BtB transcript]	**DRIVETRAIN** Swerve (auto-shoot puts the swerve in X-mode to resist defense) [BtB transcript]
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Custom CyberKraken power/CAN boards; auto routines not detailed.
VISION FOUR Arducams -> a Mac Mini running PhotonVision; if any camera sees 2 AprilTags it fuses with odometry for full field awareness. Runs ROS 2 (a node per subsystem, so one crash doesn't take down the robot) + Molex locking connectors; full match logging (all node messages + 4 camera feeds) for replay [BtB transcript]	**PUBLIC CAD / DESIGN** Full CAD + code + docs (Onshape + SolidWorks)

LESSONS FOR STUDENTS

- Diagnose and fix recurring failures: they cured belt-snap by swapping to chain at the highest-load point.
- Custom 'CyberKraken' power/CAN boards with Anderson connectors (chosen over Molex SL for shock resistance) - real electrical-design depth.
- Win by refining a proven architecture through continuous in-season optimization.
- Public ROS-based robot code is an advanced software study target.

CAD release: https://www.chiefdelphi.com/t/team-195-the-cyberknights-2026-cad-code-and-documentation-release/521390 · CD CAD/code/docs release · Onshape CAD · Behind the Bumpers · GitHub

#79 6328 Mechanical Advantage Darwin

Littleton, MA

EPA 204

51-13

Software-first team: reliable turret shooter built around AdvantageKit + Northstar vision

SHOOTER Turret shooter with adjustable 3D-printed (PLA+) hood; 4in WCP Fairlane main wheel + 2in solid-core flex top wheel on a live single shaft; optional brass flywheel for added inertia; hood gear routed from WCP SRPP	**INTAKE** Compliant-wheel roller intake (prototyped publicly) feeding a centering/serializing path
INDEXER / HOPPER Intake -> centering hopper -> tower uptake -> turret/flywheel; serialization handled in software so balls feed one at a time	**DRIVETRAIN** Swerve; .090/.063 5052 aluminum sheet with lightening tricks
ENDGAME Climber not detailed in the build thread.	**AUTONOMOUS** Software-first: AdvantageKit + Northstar AprilTag localization (Orange Pi); strong auto-aim / autonomous pedigree (routines in the released code).
VISION 'Northstar' custom AprilTag pipeline on Orange Pi coprocessors; alpha camera mounted behind the shooter hood for turret-to-hub alignment; full AdvantageKit/AdvantageScope logging + sim	**PUBLIC CAD / DESIGN** Open + code (AdvantageKit); CAD links in build thread

LESSONS FOR STUDENTS

- AdvantageKit IO-layer architecture and deterministic input->output ordering - a gold-standard software study.
- Northstar AprilTag localization on Orange Pi is a buildable, open-source vision stack.
- Routing custom gear teeth (SRPP/10DP) for turret and hood - a manufacturing technique they openly teach.
- Live-axle shooter packaging and brass-flywheel inertia trade-offs.

Code (GitHub): https://github.com/Mechanical-Advantage/RobotCode2026Public · CD build thread · Reveal 'Darwin' · GitHub (public code)

#80 3255 SuperNURDs The Thing

California, USA

EPA 202

37-23

'The Thing': 1678/254-style 'big dumper' high-flywheel FUEL scorer (rebuilt to this archetype in week 2); steel-drum flywheel with aluminum kicker/hood rollers; fully state-machine control architecture

SHOOTER Big-dumper flywheel scorer. Deliberately does NOT use an inertia (mass) wheel; instead a 4 in steel drum serves as the flywheel, storing energy in the drum mass itself, with 2 in aluminum drums acting as the kicker/hood rollers that meter and launch FUEL. The team converted to this 1678/254 'big dumper' layout mid-season (week 2), trading their original design for the proven high-flywheel dumper.	**INTAKE** unconfirmed from the release post (image + summary only) - big-dumper archetype implies a wide floor/roller intake feeding the elevated dumper; see the GrabCAD assembly for the intake and feed geometry.
INDEXER / HOPPER unconfirmed from the post - inspect the GrabCAD model for the hopper/indexer path into the dumper flywheel.	**DRIVETRAIN** Swerve (module specifics unconfirmed in the post). Notable control feature: 'Datum' anti-defense hardstops that let the robot quickly align to the hub, brace against the trench, and resist being pushed.
ENDGAME	**AUTONOMOUS**
VISION 3x Limelight 4.	**PUBLIC CAD / DESIGN** CAD release (GrabCAD)

LESSONS FOR STUDENTS

- How a mid-season pivot to the elite 1678/254 'big dumper' archetype (in week 2) can rescue and elevate a season to a top-100 world finish.
- A steel-drum flywheel (mass in the drum) as an alternative to a dedicated inertia wheel for a high-dumper shooter.
- 'Datum' anti-defense alignment hardstops: mechanical features that snap the robot into a known pose at the hub / trench to speed alignment and resist defense.
- A fully state-machine-based control architecture for a scoring robot.

GrabCAD: https://grabcad.com/library/frc-2026-team-3255-the-supernurds-1 · CD - 2026 CAD Release (Team 3255) · GrabCAD - The SuperNURDs 2026 · Statbotics 3255 (2026) · The Blue Alliance 3255 (2026)

#87 (study pick)	**1156 Under Control** Tempest Brazil	**EPA 198** 48-12

'Tempest': high-capacity swerve turret shooter — intake → hopper → spindexer → feeder → ±250° turret → adjustable-hood flywheel, plus a climber

SHOOTER Turret-mounted adjustable-hood flywheel. Hood 15–55° via rack (1× NEO 2.0, 15.37:1 MAX Planetary). Flywheel: 2× Kraken X60 (1.4:1) driving 2× brass inertia flywheels (~150 g ea) + 2× 4 in urethane (WCP 60A). Turret: 6.5 in X-contact bearing, 42.24:1 (96T/10T), 1× Kraken X44, ±250°. Shoot-on-the-move via calibrated interpolation maps (distance → hood angle / RPM / flight time), 20 iterations per cycle.	**INTAKE** 'Touch-it-own-it' ground intake. 8 mm 7075 towers + 1/2 in polycarb arms (10DP rack); front 2.25 in (AM 35A) + rear 1.25 in rollers driven by 2× NEO Vortex (2:1, ~33 ft/s). Rack expansion by 1× NEO Vortex (5:1) deploys in under 1 s; 45° polycarb ramp.
INDEXER / HOPPER Hopper → Spindexer → Feeder. Hopper: polycarbonate + carbon-fiber, expands ~12 in to hold up to ~55 FUELS. Spindexer: 3D-printed cone, 2× NEO Vortex (4:1), keeps pieces moving (anti-jam). Feeder: 1× Kraken X60 (2.2:1), 16 compliant rollers, multi-piece flow with no dead zones.	**DRIVETRAIN** Custom SDS MK4n swerve on a 28×26 in frame (2×1 in Al tube, 2 mm bellypan). L1+ (14.2 ft/s); Kraken X60 drive + Kraken X44 steer (field-oriented); machined hubs with TPU treads for grip.
ENDGAME Climber on the turret shooter (binder-documented).	**AUTONOMOUS** Shoot-on-the-move via calibrated interpolation maps (distance->hood/RPM/flight-time), 20 iterations per cycle; 4-camera AprilTag fusion.
VISION 4-camera fusion: 2× Limelight + 2× Arducam → MultiCameraPoseEstimation with AprilTag corrections feeding odometry. Turret angle solved by trig + Chinese Remainder Theorem on two absolute encoders for a 720° usable range.	**PUBLIC CAD / DESIGN** Full CAD (Onshape) + 32-page technical binder

LESSONS FOR STUDENTS

- A 3DM (data-driven) process drove the design: priority list + archetype decision matrix + functional requirements (e.g. <22.25 in to clear the trench, <21 kg modules for air travel).
- The binder documents real motor / ratio / material callouts for every subsystem — a model for a complete technical binder.
- Chinese Remainder Theorem on two absolute encoders yields a 720° turret range without re-zeroing — an elegant controls trick.
- Shoot-on-the-move uses experimentally-calibrated interpolation maps (not real-time physics) iterated 20×/cycle — a practical approach to moving shots.

Onshape CAD: https://cad.onshape.com/documents/96bff94aa06158572d9954cd/w/156a28a4c439f1f7e5f3bf04/e/3a4d945dab8b9e666f1c061c · CD CAD + tech binder release · 32-page technical binder (Drive) · Statbotics

#93 (study pick)	**5987 Galaxia (i.m.o. David Zohar)** (2026 comp bot) Petah Tikva, Israel	**EPA 197** 6-6

Turret-on-hood shooter fed by a spindexer; physics-model firing solutions

SHOOTER Turret-mounted flywheel + adjustable hood; firing solution (wheel speed, turret/hood angle) from a physics model (air resistance, Magnus force) instead of a lookup table; auto-opens hood after trench from pose/velocity; Kraken X44; SOTM in progress	**INTAKE** Floor intake on revised roller geometry tuned for rapid multi-ball pickup without jamming; used for center-field steals in auto
INDEXER / HOPPER Spindexer -> curved ramp -> gate/kicker wheels; HDPE hopper walls (replaced polycarb) to reduce sticking; counter-rotating mecanum agitator (paintball-hopper inspired) forces balls down; 2x CANrange sensors; ~55-ball goal	**DRIVETRAIN** WCP SwerveX2c; Pigeon 2.0 IMU; PathPlanner autos for neutral-zone steals
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Physics-model firing solution (wheel speed + turret/hood angle from an air-resistance & Magnus model); hood auto-opens after the trench from pose/velocity.
VISION PhotonVision on Orange Pi with AprilTags around the central target; local 3D pose algorithm	**PUBLIC CAD / DESIGN** Full Onshape CAD (Open Alliance)

LESSONS FOR STUDENTS

- A physics-based firing model beats a lookup table - re-tuning means changing a few parameters, not re-populating a table.
- Material choice fixes jams: swapping polycarb hopper walls to slipperier HDPE reduced sticking.
- Borrow cross-domain ideas - paintball counter-rotating agitators inspired their mecanum de-jam wheel.
- Program in simulation first so new students can write/tune code before the robot exists.

CD thread: https://www.chiefdelphi.com/t/frc-5987-galaxia-2026-build-thread-open-alliance/505340 · CD build thread · In-season subsystem post · GitHub

#96	**5406 Celt-X** CX26 Hamilton, ON	**EPA 196** 45-15

Strategy-first: wide touch-it floor intake + flow-through hopper + turret single-file shooter

SHOOTER Turret-mounted flywheel, single-file feed; ~8 balls/sec peak for ~2s then ~5 sustained; re-geared flywheel pulleys (15:30 -> 22:24) for higher speed; used motion-profile / current control (6328-inspired) to tame large startup PID error	**INTAKE** Wide touch-it-own-it floor intake (3 balls wide); ran ~50% speed to avoid ejecting balls over the hopper; also acts as a compactor
INDEXER / HOPPER Flow-through hopper (~45+ balls) with spindexer-style feed; throughput drops as far-from-spindexer balls settle slowly; adding agitators	**DRIVETRAIN** Swerve; emphasis on under-trench driveability + fast cycles vs double-wide shooters
ENDGAME No climber - their strategy analysis flagged climbing as low-value.	**AUTONOMOUS** Strategy-first; auto routines not detailed.
VISION Unconfirmed (AprilTag targeting expected; not detailed)	**PUBLIC CAD / DESIGN** Onshape CX26 assembly (Open Alliance)

LESSONS FOR STUDENTS

- Build an Alpha bot to surface integration problems early (melting sidewalls, chain backlash, hopper dead zones) before the comp design.
- Know your shooter's real throughput CURVE - peak BPS means little if it collapses once near-spindexer balls are gone.
- Flywheel PID from a dead stop oscillates; a motion profile / constant-current model tames startup error.
- Single-wide shooters lose a measured throughput race to double-wide; compensate with turret + driveability + faster cycles.

CD thread: https://www.chiefdelphi.com/t/celt-x-5406-2026-build-thread-open-alliance/511200 · CD build thread · Alpha-bot throughput analysis · GitHub

#100 (study pick)	**971 Spartan Robotics** Mixtape Mountain View, CA	**EPA 196** 36-28

'Mixtape': double-turret flywheel shooter (two shooters + spindexer) w/ lead-screw adjustable hoods; swerve

SHOOTER DOUBLE turret - two flywheel shooters, each with an adjustable hood driven by a lead screw. The lead screw was chosen to minimize the turret's swept radius: it tucks into the empty 'sides' instead of making the shooter longer. One iteration let the screw creep upward and lock the hood down; fixed by tapping and bolting the bottom of the screw (fix not shown in CAD). Wheel/motor specifics: read from the Onshape model.	**INTAKE** Ground intake feeding the hopper/spindexer (specifics not stated in the CD thread - see CAD).
INDEXER / HOPPER Simple spindexer feeding both shooters. Team reasoning: at high BPS it is easier to index into TWO shooters than one; they considered reviving their famous 2017 dye rotor (similar ball-shooting game) but judged it too finicky after losing the mentor talent behind the 2017 design.	**DRIVETRAIN** Swerve (module and gearing not stated in the thread - see CAD).
ENDGAME Climber not detailed (first public CAD year).	**AUTONOMOUS** Double-turret architecture; 2026 code in Java (a break from the historic C++ stack); auto routines in the released code.
VISION Custom in-house vision pipeline (971 tradition); 2026 hardware unconfirmed.	**PUBLIC CAD / DESIGN** Full robot CAD (Onshape) + code (GitHub, frc971/971-second-robot-2026)

LESSONS FOR STUDENTS

- A double turret converts a hard indexing problem into an easier one: two simple feed paths at moderate rate instead of one path at very high rate.
- Lead-screw hood actuation as a packaging move - use the dead space beside the turret to shrink swept radius rather than growing the shooter envelope.
- Institutional knowledge is a real design constraint: they rejected their own well-known 2017 dye rotor because the people who understood it were gone.
- First year of public 971 CAD, and 2026 code in Java (a break from their historic in-house C++ stack) - a newly study-able, established program.

Onshape CAD: https://frc971.onshape.com/documents/cabaa0c1c77517916df80783/w/48cb057db38a03cc202ff44e/e/96844befd4591dac162d657b · CAD + Code release (CD) · CAD (Onshape) · Code (GitHub) · Statbotics

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#103 (study pick)	**341 Miss Daisy** Miss Daisy XXIV Ambler, PA	**EPA 194** 52-9

Hopperless 360+ turret shooter with a 4-ball-wide intake and a direct 4-2-1 ball path

SHOOTER Flywheel + fully adjustable hood (team's first) on a 360+ degree turret; fed straight from intake to shooter (no storage hopper)	**INTAKE** 4-ball-wide bottom roller intake
INDEXER / HOPPER No hopper - a '4-2-1' path narrows 4 lanes to 2 to 1 directly into the turret; ~8-ball working capacity	**DRIVETRAIN** WCP X2i swerve, 27in square base, 115 lb
ENDGAME Lightweight L1 climber kept deliberately cheap so mass/time goes to scoring.	**AUTONOMOUS** 360+ turret + adjustable hood for auto-aim; the hopperless 4-2-1 feed supports a fast continuous cycle.
VISION Single turret-mounted Limelight ('limelight-turret'), MegaTag2, IMU-fused; PhotonVision IO also present; ~8.25in turret-relative camera offset	**PUBLIC CAD / DESIGN** Full CAD + code + binder + pre-match checklists

LESSONS FOR STUDENTS

- Going 'hopperless' with a continuous 4-2-1 narrowing path removes a storage subsystem and its jam points - viable if intake-to-shoot rate is high.
- An adjustable hood lets one flywheel cover many distances; pair with turret+shooter code so aiming is automatic.
- Reliability is a process: they publish per-match pit checklists and run a role-specialized crew.
- Keep the climber cheap (lightweight L1) so mass/time goes to the scoring mechanisms that drive EPA.

CAD release: https://www.chiefdelphi.com/t/team-341-miss-daisy-2026-cad-code-and-documentation-release/521088 · CD CAD/code/binder release · Reveal (Miss Daisy XXIV) · Robot code (public)

#123 (study pick)	**9072 TigerBots** Asura + Sandstorm Chesapeake, MD	**EPA 186** 52-18

Two-robot program: 'Asura' dual 720-deg twin-turret shooter, and 'Sandstorm' a 254/1678-style 4-wide dumper

SHOOTER Asura: TWO independent 720-degree turrets, each its own flywheel (~10 BPS combined). Sandstorm: 4-ball-wide top-level dumper/shooter (~19 BPS). Motor packaging iterated across test bots	**INTAKE** Full-width roller intake feeding the indexer; intake-deploy gearbox added last
INDEXER / HOPPER Asura: ~45-ball hopper. Sandstorm: ~75-ball hopper. Concluded early they'd be 'indexer-limited' and hunted the fastest indexer concepts	**DRIVETRAIN** CTRE swerve, Kraken X60 drive+steer, CANcoder, 7.67:1, 1.95in wheels, ~4.0 m/s
ENDGAME Climber not detailed (two-robot program).	**AUTONOMOUS** Concluded indexing throughput was the binding constraint; auto routines not detailed.
VISION 3x Limelight (left/center/right) via MegaTag	**PUBLIC CAD / DESIGN** CAD + code for BOTH robots (Onshape/GitHub)

LESSONS FOR STUDENTS

- Prototype the game piece FAST and find your binding constraint: they proved intake/shoot were easy and that INDEXING throughput was the real limit.
- Mine prior art - they studied 254 & 1678 plus RI3D/Open Alliance before committing.
- A dual-720-turret robot is buildable by a young team if you stage it (test bot first, then turret controls + tensioning).
- Same-team A/B study: directly compare their dual-turret vs dumper design choices.

CAD release: https://www.chiefdelphi.com/t/team-9072-tigerbots-2026-cad-and-code-release/521421 · CD CAD + code release (both bots) · Team robot page · Robot code (public)

#125 (study pick)	**2767 Stryke Force** (2026 comp bot) Kalamazoo, MI	**EPA 186** 55-12

High-quality fabrication: single shoot-on-the-move turret + spindexer on a DIY swerve (82% win rate)

SHOOTER Single turret with a 'breadbox' adjustable hood; 3D-printed turret bearing; inverted motor mount; built for shoot-on-the-move	**INTAKE** Roller intake (hardened States->Worlds; no major redesign)
INDEXER / HOPPER Spindexer feeding the turret; intentionally flexes out of the way for battery changes (ergonomics); no drum	**DRIVETRAIN** DIY in-house 'Third Coast' swerve on a Chrome-Moly frame; many parts 3D printed
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Built for shoot-on-the-move (single 'breadbox'-hood turret); fast-iterated vision.
VISION FOUR cameras (two on each corner) on TWO Orange Pis (tucked under the batteries) running their custom 'Walleye' AprilTag software (in use since AprilTags launched); used heavily to correct the gyro/odometry in this odometry-driven game [BtB transcript]	**PUBLIC CAD / DESIGN** Onshape (2026 robot)

LESSONS FOR STUDENTS

- DIY swerve design - their full mechanical design doc (Third Coast) is public.
- 3D-printed turret bearing + inverted motor mount - clever, copyable packaging.
- A spindexer that flexes for serviceability - design for repair / battery access.
- Iterate for robustness (intake hardening) without scope creep.

Onshape CAD: https://cad.onshape.com/documents/a4cd67d32bba9211459890b2 · CD 2026 CAD · Onshape CAD · Third Coast swerve design doc · GitHub · Behind the Bumpers (full transcript)

#127	**360 The Revolution** Rainmaker XXVI Tacoma, WA	**EPA 186** 40-20

Full-width turret shooter, bang-bang flywheel, shoot-on-move; 254/2910-style code

SHOOTER Full-width turret; 6328-style bang-bang flywheel wrapped in a state machine; shot interpolation from data; SOTM capped ~0.5 m/s (rising); time-of-flight tracking; passing/shooting auto-selected by robot position + match time	**INTAKE** Independent intake state machine (full-width feed); mechanical detail not fully specified
INDEXER / HOPPER Feeds the turret; managed within the superstructure state machine	**DRIVETRAIN** Swerve; PathPlanner autos (kept partly command-based to avoid state-machine conflicts)
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Shoot-on-the-move (~0.5 m/s, rising) with time-of-flight tracking; passing vs shooting auto-selected by robot position; 254/2910-style state machine.
VISION Limelight + AprilTags (AprilTag field layout; a lazy-init bug once cost a loop overrun + ~5 hrs at field)	**PUBLIC CAD / DESIGN** Onshape CAD + public GitHub

LESSONS FOR STUDENTS

- Superstructure/state-machine control (a la 2910/254 ControlState) is more flexible & testable than scattered commands.
- Adopt proven patterns wholesale: 6328 bang-bang + AdvantageKit IO layer, 254 ControlState for isolating subsystems to test.
- A CI pipeline (build/format/sim gates) + an AI first-pass review scales code quality - caught ~35 bugs incl. a missing switch break that would have broken aiming.
- The real bottleneck is PR review / learning / robot test time, not code generation.

Onshape CAD: https://cad.onshape.com/documents/91500087e03b9e406a527d6d · CD build thread · Onshape CAD · GitHub RainMaker26

#158 (study pick)	**190 Gompei and the HERD (WPI)** V1 'Doomspiral' + V2 'TURNOVER' Worcester, MA	**EPA 176** 78-26

College-mentored team that built and CAD-released TWO distinct competition robots

SHOOTER V2 TURNOVER is a turret shooter with a belt 'kicker' that feeds/compresses the ball into the flywheel (they stayed turret while many moved to drums)	**INTAKE** Kraken X60 FOC roller intake (integrated 'turnover' architecture)
INDEXER / HOPPER Kraken X60 FOC rollers route fuel through the 'turnover' path into the turret	**DRIVETRAIN** CTRE swerve, Kraken X60 FOC
ENDGAME Climber not detailed (two robots: V1 'Doomspiral' + V2 'TURNOVER').	**AUTONOMOUS** Auto routines not detailed; the study value is the V1-vs-V2 A/B comparison.
VISION 3x Limelight 4 (shooter-mounted, climber, intake positions), MegaTag2 via their GompeiLib	**PUBLIC CAD / DESIGN** Onshape - both robots (Doomspiral V1 + TURNOVER V2)

LESSONS FOR STUDENTS

- Comparing two of one team's own robots (V1 vs V2) is a rare A/B design study - what changed and why.
- Turret-vs-drum trade study: they went turret while the field trended to drums - good design-decision reasoning.
- A belt-kicker that compresses the ball into the flywheel is a clean, copyable feeding concept.

Onshape CAD: https://frc190.onshape.com/documents/ba8a365713d40fc24cf0b54f · CD 190 CAD release (both robots) · TURNOVER showcase · Onshape (TURNOVER) · Robot code (2k26)

#153 (study pick)	7415 Jaguar Robotics California, USA	EPA 176 49-17

Swappable-hopper quad-lane shooter ('Noah's Ark'): tall ~100-fuel config or short trench config, swapped in <5 min

SHOOTER Fixed (non-turret) QUAD launcher - four lanes, each on its own Kraken X60, spin-up under 0.25 s; kept under 22 in tall to pass under the trench.	**INTAKE** 'Lintake' - a 5+ fuel-wide LINEAR intake with a powered kicker roller.
INDEXER / HOPPER Swappable hopper (~100-fuel tall config OR a short trench config, swapped in under 5 min via 8 bolts) feeding six powered floor rollers with zero dead zones.	**DRIVETRAIN** 23.5 x 31.5 in base on SDS MK5n swerve (geared to R2), 7075 punch tube.
ENDGAME Designed around scoring rather than climbing the Tower (climber not a focus).	**AUTONOMOUS** Full-field localization + shooter automation for scoring autos; PathPlanner-based routines (binder details the autonomous + shot-automation sections).
VISION 4x Limelight 4 (LL4) full-field localization.	**PUBLIC CAD / DESIGN** Full CAD (Onshape) + code (GitHub) + web technical binder

LESSONS FOR STUDENTS

- A swappable hopper that reconfigures tall-vs-trench between matches exploits the multi-configuration inspection rules - capacity when you can, low profile to clear the trench and beat defense.
- An 'Alphabot' pre-season testbed surfaced problems early (inaccurate drum shot, illegal start config) before the real build.
- Four independent shooter lanes (one Kraken X60 each) give a high fire rate with sub-0.25 s spin-up.

Onshape CAD: https://cad.onshape.com/documents/d7e50bc5f2f3accc3e502b3e/w/6c7c691839b17298be38e609/e/a406643d94e0f4b9284c165c · CAD/Code/Binder release (CD) · Technical Binder · CAD (Onshape) · Code (GitHub) · Statbotics

#224 (study pick)	**3847 Spectrum** Infrared (FM) Texas, USA	**EPA 158** 39-27

'Infrared': turreted hood launcher, extending roller-bed intake, ball-tower indexer; 5-robot dev line (AM/XM/PM/FM/Photon); swerve

SHOOTER Turret-mounted launcher with hood. Per the blog motor budget: turret on a Kraken X44, 2x Kraken X60 launcher flywheel, hood on an X44 or servos; Limelight rides on the turret (visible in the FM CAD next to a Talon FX-controlled motor).	**INTAKE** 12 in extending roller-bed intake (XM sketches): rack-gear extension (linear or slightly curved), single 2 in roller with a fold-down kicker bar, deploy motors stationary near the front frame rail. Final geometry sits 5 deg over the swerve module - the smallest angle that still compresses FUEL into the carpet (per mentor Q&A).
INDEXER / HOPPER Roller bed feeds two-ball inputs through 2-to-1 gates into a single ball tower that feeds the turret; lower powered rollers plus top rollers vector the balls. Hopper walls are laser-cut polypropylene (used as a laserable, Loctite-safe polycarb substitute). They also published a 4-lane 'sideways spindexer' concept (4x throughput idea) they considered for the rebuild.	**DRIVETRAIN** Swerve. AM (alpha) ran an existing MK4i base with MK5n considered for space claims; later machines refined the same footprint (final module not explicitly stated in the blog).
ENDGAME Climber not detailed in the released dev-line CAD.	**AUTONOMOUS** Dev-line approach (AM/XM/PM/FM): final vision/auto decisions deliberately deferred until after seeing the game at real events.
VISION Limelight mounted on the turret; blog discusses aiming via odometry vs turret-camera tx/ty, with AprilTags across the field providing pose updates.	**PUBLIC CAD / DESIGN** CAD for all 5 robots (Onshape)

LESSONS FOR STUDENTS

- Their AM/XM/PM/FM/Photon 5-robot cadence intentionally delays final design decisions until AFTER seeing the game played at real events - architecture as a season-long iteration, not a kickoff bet.
- Publishing every machine's CAD (alpha through final) gives a rare public record of how one architecture matures across four rebuilds in a single season.
- Small geometric insights matter: the intake angle is exactly the minimum (5 deg) that clears the swerve module while still pressing FUEL into the carpet.
- Material pragmatism: polypropylene hopper walls because they laser-cut safely (polycarb doesn't) and take Loctite - process constraints driving material choice.

CAD release: http://2026cad.spectrum3847.org/ · OA Build Blog (CD, 416 posts) · 2026 CAD - all 5 robots · Statbotics

#232 **1756 Argos** Miami
Illinois, USA

EPA 155
32-17

'Miami': MK5n swerve, Madtown(1323)-style roller intake v2, short spindexer indexer, rear Limelight mount; launcher detail in CAD

SHOOTER Launcher specifics not documented anywhere public - the top-rear roller launcher assembly is visible in the Fusion model; read ratios/motors from the CAD.	**INTAKE** 'Madtown_Intake_v2' (92-part subassembly): a roller ground intake in the style popularized by 1323 Madtown Madness, iterated to v146 in their Fusion history.
INDEXER / HOPPER 'Spindex_Short' - a compact spindexer, by far the largest subassembly in the model (397 parts). Combined with the intake naming, the architecture is intake -> short spindexer -> launcher.	**DRIVETRAIN** SDS MK5n swerve base ('Swerve_Base_mk5n_2026', 93 parts) on a sheet-metal chassis with riveted side panels.
ENDGAME Climber not documented in the public CAD.	**AUTONOMOUS** Rear Limelight mount modeled for vision coverage; auto routines not public.
VISION Limelight on a dedicated rear mount ('back_LimeLight_Mount'); two camera-FOV volumes ('FOV Off') are modeled in the assembly to plan vision coverage.	**PUBLIC CAD / DESIGN** Full robot CAD (Fusion 360 / A360)

LESSONS FOR STUDENTS

- Modeling camera FOV volumes directly in the robot assembly is a cheap way to verify vision coverage before building anything.
- Naming a subsystem after its inspiration ('Madtown intake') and iterating it 140+ versions shows healthy borrow-then-refine engineering.
- A 'short' spindexer keeps the proven anti-jam architecture while reclaiming vertical space - worth comparing against full-height spindexers like 1756's previous seasons or 3847's concept studies.
- A complete, well-organized Fusion 360 release is rare (most teams publish Onshape) - valuable for Fusion-based teams wanting a reference structure.

Team site: https://a360.co/4c8kxxw · Full CAD (Fusion/A360) · Spectrum CAD Collection · Statbotics

#252	**1714 MORE Robotics** Steel (rebuild of Flint) Wisconsin, USA	**EPA 150** 43-20

'Steel' (mid-season rebuild of 'Flint'): 17 in-wide rear drum shooter with adjustable hood, 4x Kraken X44; roller-floor intake; deployable polycarb blocker; only the drivetrain and intake pivot carried over from Flint.

SHOOTER 17 in-wide rear-mounted drum shooter with an adjustable hood, powered by 4x Kraken X44. The team chose X44 over X60 for power density (W/lb) because making weight drove the rebuild: they budgeted ~200 A for a shot (~120 A to the shooter, ~80 A to rollers/indexer), and X44/X60 give similar peak power at a 40 A limit while four X44s save ~2 lb. The earlier robot 'Flint' used a single-wide FRONT fixed shooter (2x Kraken X60, later 2x X44) with an adjustable hood. Drum diameter, wheel/hood geometry and gear ratios beyond these are not stated in the thread - read them from the Onshape model.	**INTAKE** Roller-floor intake; the intake pivot was one of only two subsystems carried over from Flint. Many rollers/pulleys are 3D printed (eSun PLA+, some PETG-CF/PA-CF). Exact roller count/geometry not stated - inspect the Onshape CAD.
INDEXER / HOPPER Hopper/indexer feeding the drum shooter; the front of the hopper is attached to the intake so they fold down together, and it was cut down significantly to fit in frame and respect extension limits. Detailed path geometry not stated - see the CAD.	**DRIVETRAIN** Carried over from Flint. Drivetrain type is not stated in the thread - unconfirmed (do not assume); see the Onshape CAD.
ENDGAME A deployable polycarbonate blocker (defensive) doubles as the folded-down front of the hopper when stowed; it was redone between Worlds and MROC after shearing the poly in Curie Finals 2, widened, and held down by surgical tubing looped to the bash bar when deployed. Climber/traversal endgame not described - unconfirmed.	**AUTONOMOUS** Won an Autonomous Award at the Mukwonago district event; specific routines are not enumerated in the thread.
VISION Not mentioned in the release thread - unconfirmed; check the CAD/config.	**PUBLIC CAD / DESIGN** 2026 CAD (Onshape, MROC config)

LESSONS FOR STUDENTS

- A textbook REBUILT case study: a district team rebuilt nearly the entire robot mid-season (fixed front shooter 'Flint' -> 17 in rear drum shooter 'Steel'), keeping only the drivetrain and intake pivot, then won the Curie Division at Worlds.
- Motor selection driven by weight: 4x Kraken X44 on the shooter instead of X60 for power density - similar peak power at a 40 A limit but ~2 lb lighter across four motors, decisive on a weight-constrained rebuild.
- Current budgeting up front: ~200 A total per shot, split ~120 A shooter / ~80 A rollers+indexer, sized to avoid brownouts.
- Integrating defense into structure: a deployable poly blocker that also serves as the folded-down front of the hopper; iterated (widened, surgical-tubing retention) after it sheared under match loads.
- Practical 3D-printed rollers/pulleys: eSun PLA+ (some PETG-CF/PA-CF), solid walls, large layer height, hex-bore inserts, printed upright, with exposed parts protected and swapped frequently.

Onshape CAD: https://cad.onshape.com/documents/db03fec505ded9ba473e7325 · Post-season thread + CAD (CD) · CAD (Onshape, MROC) · Statbotics 1714 (2026) · The Blue Alliance 1714 (2026)

#259 (study pick)	**5847 Ironclad** (2026 comp bot) Bradley, IL	**EPA 149** 31-13

First-year Open Alliance team that won the Bluff Country Regional with turret + spindexer + adjustable hood

SHOOTER Hooded shooter with an adjustable hood; the hood was a key iteration point - broke two before settling on a wide-bottomed, more rigid design	**INTAKE** Roller intake (prototyped alongside shooter/spindexer/hopper using KrayonCAD concepting)
INDEXER / HOPPER Spindexer feeding the shooter, plus a hopper; tuned in code for spindexer rotation/feeding	**DRIVETRAIN** Swerve, Kraken X60 drive + Kraken X44 steer, top-mount 'brainpan' electronics; built to shoot-on-the-move
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Auto routines not detailed (first-year Open Alliance Regional winner).
VISION Limelight for turret aiming + pose; code adds voltage/supply/torque inputs and tunes intake, feed, shoot, spindexer, hood angle, and turret position	**PUBLIC CAD / DESIGN** Onshape (2026 robot)

LESSONS FOR STUDENTS

- A hood failure-and-redesign cycle (broke two, then widened the base) - a concrete structural-iteration lesson.
- Design prototypes for adjustability/versatility from one set of core parts (sliding-hood compression plate).
- Limelight-based turret aiming with code-tuned mechanism setpoints - an accessible vision approach for a mid-tier team.
- A first-year Open Alliance team documenting its way to a regional win.

Onshape CAD: https://cad.onshape.com/documents/b3b80a0c9175b5a0ee25425b · CD build thread · Onshape CAD · Behind the Bumpers

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#269	**3928 Team Neutrino** Halcyon Ames, IA	**EPA 147** 26-16

Tall spindexer-turret architecture with a full-width linear (rack-and-pinion) intake; Iowa Regional winner

SHOOTER Flywheel shooter on a 360+ degree turret atop a tall spindexer tower	**INTAKE** Full-width LINEAR intake driven by a 10-DP rack and pinion (tooth profile modified to cut with a 1/8in end mill, inspired by 3467/6328); racks went polycarb -> 2x .2in MAXComposite + 1x .1in aluminum for tooth strength
INDEXER / HOPPER Tall spindexer feeding the turret; ~80-fuel hopper	**DRIVETRAIN** Swerve (module unconfirmed)
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Auto routines not detailed (Iowa Regional winner).
VISION Unconfirmed (turret-aim code released)	**PUBLIC CAD / DESIGN** Full CAD release (Onshape) - first comp robot fully in Onshape

LESSONS FOR STUDENTS

- A rack-and-pinion linear intake gives a controlled straight-line deploy; 10-DP teeth modified for a 1/8in end mill make them CNC-able.
- Material selection is iterative: polycarb rack teeth sheared, so they moved to a MAXComposite + aluminum laminate - test under real load.
- Switching to Onshape pays off with parametric top-down design + FeatureScript despite transition hiccups.
- Plan for wear: turret wire sleeving wore quickly and became a per-event replacement.

CAD release: https://www.chiefdelphi.com/t/team-3928-team-neutrino-2026-cad-release/521183 · CD CAD release · Statbotics

#282 2582 PantherBots Cobra

Lufkin, TX

EPA 144

37-20

Adjustable-hood dual-motor launcher + hopper feeder; high-volume scorer; qualified Worlds

SHOOTER Adjustable-hood launcher, two flywheel motors, linear-actuator hood read by an absolute encoder; high throughput (~260 fuel/match claimed)	**INTAKE** Floor intake feeding the hopper (works with hopper to empty fuel quickly)
INDEXER / HOPPER Hopper + feeder delivering fuel to the dual-motor launcher	**DRIVETRAIN** Swerve; drivetrain assembled early; vision still in progress late-Feb
ENDGAME Side L1 climb - the CG offset from the hook needs ~0.6 friction or a dedicated gripper.	**AUTONOMOUS** Adjustable-hood dual-motor launcher with an absolute-encoder hood for repeatable shot angles; auto routines not detailed.
VISION Unconfirmed (integration in progress during build)	**PUBLIC CAD / DESIGN** Onshape CAD + GitHub (Open Alliance blog)

LESSONS FOR STUDENTS

- A linear-actuator hood with an absolute encoder gives repeatable, tunable shot angles for variable distance.
- Side L1 climbing creates a tricky moment - CG offset from the hook needs ~0.6 friction or a dedicated gripper.
- Dual launcher motors must be monitored - a single over-current motor silently halves your shooting (watch current draw).
- Document openly (GitHub/Onshape/blog) to keep iteration disciplined through a Worlds-qualifying season.

CD thread: https://www.chiefdelphi.com/t/2582-pantherbots-build-blog-open-alliance-2026/509632 · CD build blog · Reveal 'Cobra'

#323 1778 Chill Out HAILSTORM

Washington, USA

EPA 134

25-19

'HAILSTORM': compact turret shooter fed by a spindexer; motor + single absolute-encoder turret; Kotlin code; swerve

SHOOTER Turret-mounted launcher (described as a compact turret assembly). Per the thread, the team weighed a 'dye rotor' but rejected it as too complex and chose the turret/spindexer path for simplicity with a high ceiling. Flywheel motor/ratio specifics not stated in the thread - read from the Onshape model.	**INTAKE** Ground intake; the team noted a mid-season intake redesign that delayed their practice-field time. Specifics not detailed in the thread - see CAD.
INDEXER / HOPPER Spindexer feeding the turret. The team chose a spindexer over a dye rotor as the simplest solution with a high ceiling. (The cone was finished off-CAD with cat-tongue grip-tape strips, heat-shrunk to conform for extra grip.)	**DRIVETRAIN** Swerve (module/gearing not stated in the thread - see CAD).
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Turret zeroing via a motor (relative) encoder + one absolute encoder; auto routines not detailed.
VISION Not named in the thread - the turret aims via encoders rather than a stated camera; treat as Unconfirmed and check the CAD/code repo.	**PUBLIC CAD / DESIGN** Full robot CAD (Onshape) + code (GitHub, Kotlin)

LESSONS FOR STUDENTS

- Pragmatic turret zeroing: the motor (relative) encoder handles continuous tracking, and a single absolute encoder geared ~5:1 supplies a re-zero value while shooting; closest-region logic plus manual re-zero buttons cover encoder drift - a cheap alternative to a full CRT/absolute setup they couldn't tune in time.
- Archetype selection as risk management: they rejected a dye rotor for a spindexer specifically because it was the simplest path with a high performance ceiling.
- A public Kotlin codebase is a useful non-Java reference for teams exploring JVM languages.
- Recovery story worth reading: early power-consumption issues, then a hardening/tuning focus that made them one of their team's best-ever robots by DCMP.

Onshape CAD: https://cad.onshape.com/documents/c527fb2ecdd377cbb954f474/w/b89f138221c8d9223c28e242/e/c4daa58f8630b2239d8ee484 · CAD & Code release (CD) · CAD (Onshape) · Code (GitHub) · Statbotics

#359	48 Team E.L.I.T.E. xtremachen 29 Ohio	EPA 128 29-23-1

Turret spindexer shooter, fully in-house manufactured, fast-iterated vision/SOTM; Midwest Regional winner

SHOOTER Turret-mounted flywheel with shoot-on-the-move	**INTAKE** Roller intake with a quick-change roller (a bent roller is a fast pit swap); stiffened + de-jammed after the first regional
INDEXER / HOPPER Spindexer (cone geometry inspired by 4145)	**DRIVETRAIN** Swerve (module unconfirmed)
ENDGAME Climber present but iterated for reliability (they considered removing it between events).	**AUTONOMOUS** Two Limelight 4s + pose + shoot-on-the-move brought online in ~8 days between events.
VISION 2x Limelight 4 for AprilTag pose + shoot-on-the-move; stood up in ~8 days mid-season	**PUBLIC CAD / DESIGN** Full CAD release (Onshape) - first Onshape season, 100% in-house parts

LESSONS FOR STUDENTS

- Vision is a software sprint you can win late: two Limelight 4s + pose + SOTM working in ~8 days between events.
- Design for fast field repair - a quick-change intake roller turns a bend into a pit swap, not a dead robot.
- Iterate event-to-event with a clear list (stiffen intake, fix jamming, even remove the climber for reliability).
- Borrowing a subsystem detail (spindexer cone from 4145) beats reinventing it.

CAD release: https://www.chiefdelphi.com/t/frc-48-team-e-l-i-t-e-xtremachen-29-2026-cad-release/521258 · CD CAD release · Statbotics

#377	6672 Fusion Corps Bastion Texas, USA	EPA 124 20-25

'Bastion': turreted flywheel shooter with kicker feed, floor intake and indexer, on CTRE swerve; PhotonVision AprilTag targeting. Full Onshape CAD + AdvantageKit-style code released.

SHOOTER Turreted flywheel launcher. The released code exposes separate Shooter, Turret and Kicker subsystems, each driven through CTRE TalonFX controllers (Kraken/Falcon-class motors; specific motor models unconfirmed - see the CAD BOM). The dedicated Turret subsystem means the launcher rotates independently of the chassis (shoot-while-aiming); a Kicker meters game pieces into the flywheel. Flywheel wheel/hood geometry and gear ratios are not stated in the release post - read them from the Onshape model.	**INTAKE** Floor intake driven by a CTRE TalonFX controller (Intake subsystem in code). Roller/geometry detail not described in the release post - inspect the Onshape CAD.
INDEXER / HOPPER Indexer driven by a REV SPARK controller (NEO-class motor; model unconfirmed). Sits between the intake and the kicker/turret feed. Path geometry not described in text - see the CAD.	**DRIVETRAIN** Swerve. Code uses CTRE Phoenix 6 (generated TunerConstants, TalonFX/TalonFXS module controllers, PhoenixOdometryThread) with a selectable NavX or Pigeon2 gyro, and PathPlanner autonomous. Module make/ratios unconfirmed in the post - see the CAD.
ENDGAME No climber/endgame subsystem appears in the released code; endgame mechanism (if any) unconfirmed - check the Onshape CAD.	**AUTONOMOUS** PathPlanner-based autonomous routines (pathplanner deploy folder in the repo); specific paths/sweeps not enumerated in the release post.
VISION PhotonVision for AprilTag localization (VisionIOPhotonVision + photonlib vendordep in the released code). Camera count/placement not stated in the post - see the CAD/config.	**PUBLIC CAD / DESIGN** 2026 CAD Release (Onshape)

LESSONS FOR STUDENTS

- A clean, readable AdvantageKit-style codebase: every mechanism (drive, shooter, turret, kicker, intake, indexer, vision) is split into a subsystem + hardware IO interface + sim implementation - an excellent template for teams learning that architecture.
- A separate Turret subsystem decoupled from the drivetrain shows how to structure code for a shoot-on-the-move turreted launcher.
- PhotonVision + Phoenix 6 swerve + PathPlanner is a widely reusable, fully-public localization/auto stack for a mid-resource team.
- Mixed motor-controller ecosystem in one robot (CTRE TalonFX on most mechanisms, REV SPARK on the indexer) - a realistic example of using the right controller per mechanism.

Onshape CAD: https://cad.onshape.com/documents/c8f7278d84c0d231b4ae782c/w/32ba4de6819aba4f14857523/e/4f4804f7c4c66c747eefa67f · CAD + Code release (CD) · CAD (Onshape) · Code (GitHub) · Statbotics 6672 (2026) · The Blue Alliance 6672 (2026)

#399	1676 Pascack PI-oneers Cobra Pi Montvale, NJ	EPA 123 35-24-1

High-capacity hopper shooter known for big event-to-event upgrades; ~80-ball usable storage

SHOOTER Flywheel shooter (turret detail unconfirmed); continuous upgrades each event	**INTAKE** Roller intake (motors unconfirmed); throughput slows once the hopper is heavily loaded
INDEXER / HOPPER Large hopper: ~80 balls before fuel weight bogs the intake, ~90-95 at overflow	**DRIVETRAIN** YAGSL swerve, Kraken X60 drive + Kraken X44 steer, CANcoder, navX
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Auto routines not detailed; large event-to-event upgrades.
VISION 3x Limelight (left/right/front), MegaTag pose estimation	**PUBLIC CAD / DESIGN** CAD + code release (Onshape)

LESSONS FOR STUDENTS

- Add-on Delrin 'skis' under the drivebase help the robot ride over the field bump and shield swerve motors - a cheap fix for a known hazard.
- Know your REAL capacity limit (~80 usable) vs physical (90-95) - design to the usable number.
- Aggressive event-to-event upgrades drove a steep improvement curve.

CAD release: https://www.chiefdelphi.com/t/1676-cad-release-2026/520302 · CD CAD + code release · Team mechanical page · Released code (practice chassis)

#381	2079 4-H ALARM Robotics Deadshot Franklin, MA	EPA 122 26-20

Drum shooter (no turret) with a compression-tuned over-bumper intake + roller indexer

SHOOTER Drum shooter (team very satisfied with it); planned to grind the hood for a higher-arching, more accurate shot; SOTM not yet implemented	**INTAKE** Over-bumper intake whose compression was too high - couldn't copy another team's geometry, had to run their own compression tests; broke several times and was reinforced with aluminum plates
INDEXER / HOPPER Indexer rollers feeding the drum; planned smaller indexing rollers to raise throughput	**DRIVETRAIN** Kraken X60 swerve (TalonFXS modules)
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Drum shooter; shoot-on-the-move not yet implemented (planned).
VISION 4x PhotonVision cameras (back-left, back-right, side-left, side-right) - PhotonVision, not Limelight	**PUBLIC CAD / DESIGN** Onshape CAD + GitHub (Open Alliance)

LESSONS FOR STUDENTS

- Intake compression must be tuned to YOUR robot - copying another team's spacing doesn't transfer; run your own ball-compression tests.
- Design for durability: an intake that breaks repeatedly needs structural reinforcement, not just tuning.
- Smaller indexing rollers can raise throughput - small geometry changes meaningfully affect ball flow.
- Borrow hopper-capacity ideas from top teams (passive net like 5687, vertical active expansion like 1678).

CD thread: https://www.chiefdelphi.com/t/frc-2079-alarm-robotics-2026-rebuilt-build-thread/509762 · CD build thread · GitHub · Robot code (DeadShot)

#496	4392 The Deceivers (2026 comp bot) Brimley, MI	EPA 107 24-20

Small (17-student) team that built an ambitious turret shooter with a star-wheel intake

SHOOTER Turret: dual Kraken X60 on a 1:1 shooter axle, two SDS brass flywheels + a single 4in Fairlane 60A wheel; adjustable hood with a 1in hood roller geared 2:1	**INTAKE** Over-bumper 4in compression wheels + star wheels (no kicker bar needed); deployed by dual mirrored rack-and-pinion (racks = 4 epoxied laminations of .120 polycarb, PETG pinions)
INDEXER / HOPPER Roller hopper with a passive funnel + a kicker that flicks balls into the shooter (254-2017 style); chose roller hopper over spindexer after prototyping both	**DRIVETRAIN** Swerve, Kraken X60 drive, 4.2:1, 38mm wheels, ~5.5 m/s, 0.55m track width
ENDGAME Climber not detailed (small 17-student team).	**AUTONOMOUS** Auto routines not detailed.
VISION 4x Limelight (one..four), MegaTag2	**PUBLIC CAD / DESIGN** Onshape (2026 document)

LESSONS FOR STUDENTS

- A star-wheel + compression-wheel over-bumper intake can work WITHOUT a kicker bar.
- Rack-and-pinion intake deploy with laminated-polycarb racks - fabrication on a budget.
- Roller-hopper vs spindexer: they prototyped both and documented why they picked rollers.
- Honest self-critique (3 lb of brass flywheel) models good engineering process for students.

Onshape CAD: https://cad.onshape.com/documents/589237fcf2364719e491ed4b · CD Open Alliance thread · Onshape CAD · Website · Robot code

#528	1577 Steampunk Cerberus Israel	EPA 107 5-5

Wide-intake high-capacity turret shooter, fully student-built; reached Newton Division

SHOOTER Flywheel turret shooter, ~19 BPS sustained	**INTAKE** 5-balls-wide intake
INDEXER / HOPPER Large hopper, ~75-ball capacity	**DRIVETRAIN** Swerve at an 'R2' (MAXSwerve-class) ratio; max frame perimeter, max-weight robot
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Auto routines not detailed; ~19 BPS sustained throughput.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** CAD + Technical Binder (Onshape)

LESSONS FOR STUDENTS

- A wide intake (5 across) + a big ~75-ball hopper makes a high-throughput cycle machine - width and capacity are force multipliers.
- Fully in-house, student-led manufacturing is achievable; pair it with a written technical binder so the reasoning survives to next year.
- Pick a known-good swerve ratio ('R2') instead of over-customizing, and spend the saved effort on scoring mechanisms.

CAD release: https://www.chiefdelphi.com/t/steampunk-1577-2026-cad-and-technical-binder-release/521240 · CD CAD + tech binder release · Statbotics

#710	**5811 The BONDS** Lektor Ohio	**EPA 85** 15-16

720-deg turret shooter with a 3D-printed spindexer and a 1678/27-style slapdown intake; mid-season rebuild to add a hopper

SHOOTER Flywheel on a 720-degree turret rotating on a 7in X-Contact bearing; praised for tight turret wire management + ball-exclusion shielding	**INTAKE** '27/1678-inspired' slapdown intake; later swapped to a rebuilt 'Buckeye' 4-bar intake mid-season
INDEXER / HOPPER 3D-printed spindexer; rebuilt between Wk3 and Wk5 to move the intake to the short side, freeing room to add a hopper	**DRIVETRAIN** Swerve, 32x23.5in base
ENDGAME Climber not detailed; a mid-season rebuild added a hopper.	**AUTONOMOUS** Auto routines not detailed.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** CAD release + reveal video (Onshape)

LESSONS FOR STUDENTS

- Don't fear a mid-season rebuild: moving the intake from the long side to the short side opened room for a hopper and 'significantly improved' offense.
- Borrow proven geometry - their slapdown intake is openly modeled on 1678/27, a fast path to a working mechanism.
- A 720-degree turret on a big X-contact bearing needs disciplined wiring + a ball-exclusion shield.
- 3D-printed spindexers are a low-cost, iterable storage option when you can't machine custom parts.

CAD release: https://www.chiefdelphi.com/t/team-5811-the-bonds-2026-robot-cad-release/520901 · CD CAD release · Statbotics

#730	**2183 Tigerbots** Louisiana, USA	**EPA 83** 18-18

720-deg passive central front turret; beam-break-controlled dual-roller polybelt serializer; large hopper; MK4n swerve sized to fit under the trench

SHOOTER Shooter rides a central front turret with 720 deg of travel achieved via a passive mechanism (no re-zeroing); wheel/motor specifics not stated in the thread - see CAD.	**INTAKE** Ground intake feeding the large hopper (specifics not detailed in the build thread - see CAD).
INDEXER / HOPPER Center serializer with two polybelt roller stacks that are INDEPENDENTLY controlled (not belted together) using beam-break sensors: rather than simply alternating, each roller is driven based on whether a ball is detected - sometimes both roll inward, changing behavior when a beam break trips - to de-interlock FUEL and funnel the wide hopper into one stream. (Corrected 2026-07-06 per team CAD/Code lead: earlier text wrongly assumed the two rollers were belted together and merely alternated.) Reached via 4+ prototypes - a spindexer (too sensitive to gravity/geometry) and a two-wheel track (balls interlocked) were rejected; V1 (top belt only) ran the Oklahoma and Magnolia regionals before the final version.	**DRIVETRAIN** SDS MK4n swerve; overall height constrained to drive under the trench, which motivated the low central turret architecture.
ENDGAME Climber not detailed in the public release.	**AUTONOMOUS** Auto routines not detailed.
VISION Unconfirmed (not stated in the build thread).	**PUBLIC CAD / DESIGN** Full CAD (Onshape) + subsystem build thread

LESSONS FOR STUDENTS

- Independently-driven serializer rollers with beam-break sensing (rather than one belted-together set) let the robot vary each roller's behavior by ball state to clear and prevent jams.
- Their thread shows the full failure path (spindexer -> track -> belted serializer): documenting WHY prototypes failed is as instructive as the final design.
- A passive 720-deg turret gets shoot-anywhere coverage without slip rings or complex wiring - clever low-budget engineering.
- Packaging constraints (fit under the trench with MK4n modules) drove the whole architecture - constraints first, mechanism second.

Onshape CAD: https://cad.onshape.com/documents/0f18097cc68763333bbf9187/w/ec4ab9bf696dddb823f66e6c/e/e6c97ba1f43ead9fa84d0197 · CD build thread (turret + serializer) · Full CAD (Onshape) · Statbotics

#844 (study pick)	**1601 Quantum Samurai** Linguist New York, USA	**EPA 75** 15-11

'Linguist': 360-degree turreted flywheel shooter with adjustable hood, deployable over-the-bumper rack-extension intake, tower-fed hopper/indexer; WCP X2C swerve

SHOOTER Turret-mounted. 2x Kraken X60 belted to a 4 in OD Colson wheel via shaft at 11:18; two 1.5 lb brass flywheels belted to the same Colson shaft at 18:18 to smooth and hold rotational momentum. FUEL clearance (main-plate diameter) is 6.51 in; each flywheel retained by the side plates plus 1/8 in aluminum plates. Adjustable ABS hood driven by a Kraken X44 through an ABS double-herringbone rack and pinion at >200:1 (step-up gears), giving 21-50 deg of hood travel; anti-flex pins extend into the rack, ABS e-chain manages wiring.	**INTAKE** Deployable over-the-bumper intake on the front two X2C modules. Structure: three 0.125 in polycarbonate panels joined at 90 deg with aluminum top brackets and a 1x1 0.125 in aluminum crash bar for impact. Rollers: a 2 in polycarbonate roller in silicone tubing on a 3D-printed 24T hub (WCP stub shaft) plus a 1.5 in polycarbonate roller in CatTongue grip tape on an 18T hub, belted together and to a kicker roller via a Kraken X60 at 12:18. A kicker improves FUEL compression consistency.
INDEXER / HOPPER Tower-fed hopper/indexer. Indexer = two 0.25 in polycarbonate side panels 22.5 in apart with 0.125 in polycarbonate vector backing panels; two vectored 2 in intake-roller shafts plus center compliant wheels. Before reaching the tower, FUEL is agitated by 4x 1.25 in dia aluminum hotdog rollers (20.5 in long) on 20T WCP stub rollers; all shafts belted 12:24 to a Kraken X60. The tower funnels FUEL up to the turret and is reinforced with 1/4 in polycarbonate supporting plates.	**DRIVETRAIN** 27x27 in frame on four WCP X2C swerve modules. Double-bellypan design: two aluminum bellypans mount electronics (one for the PDP, one for the RoboRIO / Pigeon IMU / CANivore), with a bottom Slippery UHMW skid panel spanning the base for debris protection and as an anti-beaching surface. 3D-printed TPU battery sock retains the battery.
ENDGAME Endgame/climber not described in the released tech binder; check the Onshape CAD.	**AUTONOMOUS** Autonomous routines not detailed in the binder; software detail lives in the team's resources.
VISION Three Limelights for full-field localization: two mounted atop the X2C modules nearest the turret tower and a third against a hopper support stock - the robot always knows its field pose for turret aiming.	**PUBLIC CAD / DESIGN** 2026 CAD + Tech Binder (Chief Delphi)

LESSONS FOR STUDENTS

- The turret is built around an X-contact bearing with a 3D-printed 184T ring gear driven by a 24T aluminum pinion (38.334:1) off a Kraken X60 / MAXPlanetary 5:1 - a fully student-documented turret others can copy.
- A two-stage flywheel (driven 4 in Colson at 11:18 plus 1.5 lb brass inertia flywheels at 18:18) is a clean way to smooth shot-to-shot energy without a heavier single rotor.
- Strong end-to-end documentation example for a mid-tier team: every one of the six mechanisms (turret, shooter, chassis, tower, hopper/indexer, intake) has a materials-and-ratios writeup plus CAD - excellent study material for newer teams.
- Process honesty: the binder credits CAD overviews from 1323 and 1678 just before their first regional, showing how design reviews from veteran teams shaped quick fixes.

CD thread: https://www.chiefdelphi.com/t/522131 · CAD + Tech Binder release (CD) · Statbotics · The Blue Alliance

#932	6381 Red Raider Robotics (2026 comp bot)	EPA 70
	Sheboygan, WI	24-23

Simplicity-first single shooter (NO turret), modular per-subsystem Onshape docs

SHOOTER Single fixed shooter (no second shooter, no turret) - chose one for accuracy, judged they can move balls fast enough; turret deferred to the offseason due to inexperience	**INTAKE** Over-bumper / floor intake (its own Onshape doc; outtake variants prototyped on YouTube)
INDEXER / HOPPER Dedicated indexer subsystem in its own Onshape doc feeding the single shooter	**DRIVETRAIN** Chassis subsystem in its own Onshape doc (swerve expected)
ENDGAME Climber not detailed (simplicity-first, no turret).	**AUTONOMOUS** Auto routines not detailed; one accurate fixed shooter by choice.
VISION Unconfirmed (no clear vision pipeline documented)	**PUBLIC CAD / DESIGN** Onshape - one doc per subsystem (Shooter/Indexer/Intake/Chassis/Climber)

LESSONS FOR STUDENTS

- Scope to your team's experience: deliberately skipping a turret keeps a developing team reliable rather than over-reaching.
- One accurate shooter can beat two if you can feed balls fast enough - fewer mechanisms, fewer failure points.
- Splitting CAD into one Onshape doc per subsystem lets students work in parallel without conflicts.
- Defer ambitious mechanisms (turret) to the offseason as a learning project rather than risking the season.

Onshape CAD: https://cad.onshape.com/documents/e139ac1c610559bc8767c8da · CD build thread · Onshape subsystem docs

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#1109	5982 Merrill's Circuits of Destruction Doubloon	EPA 60
	Michigan, USA	21-21

'Doubloon': WCP Competitive Concept-based scorer reworked for serviceability; PhotonVision

SHOOTER Based on the WCP Competitive Concept shooter (specifics not detailed in the release).	**INTAKE** Quick-change intake: every roller runs on standard hex shafts so a broken roller swaps in under a minute; geometry revised for more consistent pickup.
INDEXER / HOPPER Lightweight net hopper replacing the CC's heavy extendable hopper — lighter and more durable.	**DRIVETRAIN** Swerve, built from the WCP Competitive Concept base (module specifics unconfirmed).
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** PhotonVision; auto routines not detailed.
VISION PhotonVision — integrated for on-field targeting (not modeled in the CAD).	**PUBLIC CAD / DESIGN** Full CAD (Fusion 360)

LESSONS FOR STUDENTS

- Adapting a COTS concept (WCP Competitive Concept) and optimizing it for serviceability — quick-change hex-shaft rollers — is a legitimate, time-efficient design path.
- Replacing a complex extendable hopper with a simple net cut weight and improved durability: simplicity as a design win.

Team site: https://a360.co/3NHiaZ3 · CD CAD release · Statbotics

#1838 135 Penn Robotics (Black Knights) unconfirmed

Indiana, USA

EPA 38

22-40

Swerve, twin-turret twin-flywheel shooter; 254-2017-style floor-roller indexer, wide floor intake, omnidirectional side/center hang.

SHOOTER Twin flywheel shooters on two turrets, chosen for redundancy ('one is none, two is one' - keep firing if one jams or stalls). Each turret pairs a flywheel with an adjustable hood (about 15-45 deg hood range, turret elevation about 45-90 deg); the flywheel is either direct-driven or on a 180-degree gearbox to keep the motor parallel with the flywheel shaft. A prototype used a 4in green compliant wheel (softer neoprene-tread wheels shredded the game pieces). Motors/ratios: see the Onshape CAD.	**INTAKE** Wide floor intake - the frame was maximized on the intake/turret side for the widest possible intake. Battery and RIO were packaged under the intake to free hopper volume. Roller count/motors: see the CAD.
INDEXER / HOPPER Indexer modeled on 254's 2017 design, with team 118 also referenced. Silicone-wrapped 1in-OD 'floor rollers' geared up to keep constant upward pull on the fuel, plus 'kickup shelf' rollers near the turrets to raise balls-per-second and act as a 'jackhammer' to clear clogs. The layout is designed to stack about 3 fuel vertically across most of the robot.	**DRIVETRAIN** Swerve (the team describes 'a turret/swerve' architecture); 30x25in frame. Module brand/ratio beyond swerve is unconfirmed; the CAD has a 'DriveTrain135-2026' tab.
ENDGAME Omnidirectional hang (CAD 'Telescoping Arm' + 'Elevator'). Extending arms reach up one at a time to hang off the side or center of the ladder tower, so the robot does not block partners' side-hang spots; compliant wheels grip the bar to roll into place and reduce denting and sway. The battery is mounted under the hang to bias weight toward it.	**AUTONOMOUS** Not detailed in the thread; see the 2026-Rebuilt code repo (includes turret azimuth math).
VISION Custom machine-learning vision: their 'Southmoon' repo contains a trained model (best.pt and a best.mlpackage). The specific pipeline/product is unconfirmed - read the code.	**PUBLIC CAD / DESIGN** 2026 CAD (Onshape)

LESSONS FOR STUDENTS

- Twin-turret redundancy: 135 built two independent turret shooters so the robot can keep scoring if one fails - a concrete durability-vs-complexity trade for a scoring subsystem.
- Indexer throughput details: a 254-2017-style floor-roller indexer with silicone-wrapped 1in rollers plus 'kickup shelf' rollers near the turrets to raise balls-per-second and clear clogs.
- Omnidirectional hang: extending arms that side- or center-hang on the ladder with compliant wheels gripping the bar, designed not to block alliance partners' hang spots.
- A documentation-rich Open Alliance thread from a mid-tier team (local rank 1838, 22-40): included for the downloadable CAD and detailed design log, not for competition results. Parameterized CAD via Onshape Variable Studios. 2025 FIN Lafayette Impact Award winner.

Onshape CAD: https://cad.onshape.com/documents/20202353e5ff6b6c6fe21c64 · Open Alliance build thread (CD) · 2026 CAD (Onshape) · Code (GitHub, 2026-Rebuilt) · Statbotics 135 (2026) · The Blue Alliance 135 (2026)

#1896	157 The AZTECHS (2026 comp bot) Marlborough, MA	EPA 36 11-18

Modest on-field results but outstanding, deeply-documented vision + turret engineering

SHOOTER Two-motor single-wheel flywheel using 4in ThriftyBot urethane wheels at ~1in compression; adjustable hood 20.0-57.5 deg	**INTAKE** Roller intake; added a second Kraken X60 which greatly sped up intaking and reduced jams
INDEXER / HOPPER Polycarb-roller hopper slanted toward the uptake (grip via hockey tape / textured TPU); 254-style turret wiring; turret sensed by an effective 1:1 encoder over 360 deg	**DRIVETRAIN** Thrifty Narrow Swerve via the CTRE Swerve Project Generator; uses YAMS (Yet Another Mechanism System)
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Multi-camera PhotonVision on dual Orange Pi 5 (deeply documented vision); auto routines not detailed.
VISION PhotonVision 2026 on two Orange Pi 5 coprocessors (load split across ~4 ThriftyCams); 3 cams on the shooter platform for localization + 1 on the turret for hub alignment; custom AprilTag mounting plates	**PUBLIC CAD / DESIGN** Open Alliance build thread (CAD/links in-thread)

LESSONS FOR STUDENTS

- Multi-camera PhotonVision on dual Orange Pi 5 - splitting cameras across coprocessors, MJPEG-vs-YUYV trade-offs, per-Pi hostnames (a thorough vision study).
- Separate localization cameras (chassis) from a targeting camera (turret) - a clean architecture pattern.
- 254-style turret wiring + a 1:1 absolute encoder for a 360-deg turret.
- Adding a 2nd intake motor to cure jams - a simple, high-impact reliability fix. (Study for vision/turret, not on-field results.)

CD thread: https://www.chiefdelphi.com/t/frc-157-the-aztechs-2026-build-thread-open-alliance/509425 · CD build thread · Statbotics

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Sources: Chief Delphi (reveals & Open Alliance build threads), David's Design Server / frcdesign.org (robot-pics-2026 & mechanism-examples), team CAD/code releases (Onshape, GrabCAD, GitHub), FUN "Behind the Bumpers", and Statbotics EPA & records (2026). "Unconfirmed" means a detail was not publicly published, not that it is absent.